

Epidemic Burnout under General Susceptible Recruitment

A Unified Asymptotic Theory from Phase-Plane Matching to Second-Order x_{in} and Kendall Persistence Probability

Unified theory note

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Abstract

We develop a self-contained asymptotic framework for the probability of pathogen burnout or persistence after the first epidemic wave. We consider

$$\frac{dx}{d\sigma} = \rho(1-x)x^\theta - \mathcal{R}_0xy, \quad \frac{dy}{d\sigma} = (\mathcal{R}_0x - 1)y, \quad 0 \leq \theta \leq 1.$$

The central problem is to estimate the susceptible fraction x_{in} when the post-peak infectious fraction first falls to a low-prevalence matching line $y = y_{\text{BL}}$, without integrating the full two-dimensional epidemic ODE, and then to use that estimate to calculate the probability P_1 that the pathogen survives the first epidemic trough.

Following the matched-asymptotic phase-plane architecture of Parsons and Earn, we extend standard linear susceptible replenishment to the family $h_\theta(x) = (1-x)x^\theta$. The zero-demography epidemic geometry first gives a universal outer trajectory $F(x)$ and its final-size root x_f . We then directly expand the exact phase-plane equation in the fixed- $\mathcal{R}_0 > 1$, $\rho \rightarrow 0$ limit. The first-order correction develops a logarithmic singularity near x_f ; its finite part defines the general- θ refined matching amplitude C_θ , which reduces exactly to the refined Parsons et al. burnout prefactor when $\theta = 0$. Continuing the same expansion to $O(\rho^2)$ yields a second matching constant D_θ and the next corner switchback. Re-expressing the corner expansion in terms of the action primitive $\mathcal{A}_\theta$ gives a stable second-order action-resummed scalar equation for x_{in} .

Finally, x_{in} feeds directly into Kendall's time-inhomogeneous birth–death formula. A high-accuracy version requires only a one-dimensional Kendall integral; when the action is sufficiently large, Laplace's method gives a fully asymptotic closed form. We also distinguish several boundaries of validity: no boundary-layer entry, scale merging as $\mathcal{R}_0 \rightarrow 1^+$, small-action Laplace failure, a high- $\mathcal{R}_0$ distinguished limit, and finite- K stochastic-entry fluctuations. These are validity limits of the asymptotic regime rather than missing pieces of the main derivation.

1 Executive formula sheet: predictions, definitions, and validity conditions

This section is a self-contained statement of the result. It is intended for a reader who wants to implement or assess the approximation without following the matched-asymptotic derivation. The 2024 Parsons–Bolker–Dushoff–Earn burnout paper carries the phase-plane matching through the refined term that is called *first order* here (the constant C_0); it does not contain the D_θ second-order extension derived in this note. At second order, the outer calculation identifies D_θ as the finite part of X_2 after its known squared-logarithmic term is removed. Matching the outer and corner expansions then fixes the undetermined corner integration constant to this same D_θ .

1.1 Population-size-dependent transmission: compact extension

Section 15 supplements, rather than replaces, the validated theory below. For a fixed exponent $0 \leq \psi < 1$, put

$$\eta = 1 - \psi, \quad n = x + y, \quad F_\psi(x) = \left(1 + \frac{\eta}{\mathcal{R}_0} \log x\right)^{1/\eta} - x.$$

Its nontrivial zero is denoted by $x_{f,\psi}$ and the low-prevalence threshold is $x_{*,\psi} = \mathcal{R}_0^{-1/\eta}$. The generalized action is defined without an unnecessary special-function representation by the finite difference

$$\mathcal{A}_{\theta,\psi}(v) - \mathcal{A}_{\theta,\psi}(u) = \int_u^v \frac{1 - \mathcal{R}_0 z^\eta}{h_\theta(z)} dz. \quad (1)$$

The exact generalized Kendall integral is

$$\mathcal{I}_{\theta,\psi}(x_{\text{in}}) = \int_{x_{\text{in}}}^1 \frac{1}{\rho h_\theta(X)} \exp\left\{\frac{\mathcal{A}_{\theta,\psi}(X) - \mathcal{A}_{\theta,\psi}(x_{\text{in}})}{\rho}\right\} dX. \quad (2)$$

All matching constants and finite-height corrections are derived in Section 15. Setting $\psi = 0$ there recovers every quantity in the present formula sheet. The endpoint $\psi = 1$ is excluded.

1.2 Basic inputs and quantities derived from them

The independent dimensionless inputs are

$$\mathcal{R}_0 > 1, \quad 0 < \rho \ll 1, \quad 0 \leq \theta \leq 1, \quad K \gg 1.$$

Here $\mathcal{R}_0$ is the basic reproduction number, ρ is the susceptible recruitment rate divided by the infectious removal rate, θ specifies the recruitment law, and K is population size. Fractions of susceptible and infectious hosts are denoted by x and y . Define, in the following order,

$$\begin{aligned} h_\theta(x) &= (1 - x)x^\theta, & F(x) &= 1 - x + \frac{1}{\mathcal{R}_0} \log x, \\ x_* &= \frac{1}{\mathcal{R}_0}, & x_f &= -\frac{1}{\mathcal{R}_0} W_0(-\mathcal{R}_0 e^{-\mathcal{R}_0}). \end{aligned}$$

Thus $x_f \in (0, x_*)$ is the nontrivial root of $F(x_f) = 0$. The endemic infectious fraction and the standard matching height are

$$y^* = \rho h_\theta(x_*) = \rho(\mathcal{R}_0 - 1)\mathcal{R}_0^{-(1+\theta)}, \quad y_{\text{BL}} = \sqrt{\frac{y^*}{K}}. \quad (3)$$

The alternative choice $y_{\text{BL}} = y^*$ used in some comparison plots may be inserted in every formula below, but it is an empirical fallback: because it is $O(\rho)$ rather than $o(\rho)$, it does not have the same formal second-order error guarantee as (3).

The refined first-order constant is obtained entirely from $(\mathcal{R}_0, \theta)$:

$$\begin{aligned} \tilde{a}_\theta &= \frac{h_\theta(x_f)}{x_f}, \\ \mathcal{J}_\theta &= \int_{x_f}^1 \left[\frac{(x_* - u)h_\theta(u)}{u^2 F(u)} - \frac{\tilde{a}_\theta}{u - x_f} \right] du, \\ C_\theta &= \frac{(1 - x_f)(x_* - x_f)}{x_f} \exp\left(\frac{\mathcal{J}_\theta}{\tilde{a}_\theta}\right). \end{aligned} \quad (4)$$

The integrand in $\mathcal{J}_\theta$ has a removable endpoint singularity.

For the second-order constant, define the coefficient integrands

$$q_1(u) = -\frac{h_\theta(u)(\mathcal{R}_0 u - 1)}{\mathcal{R}_0^2 u^2 F(u)},$$

$$q_2(u) = \frac{h_\theta(u)(\mathcal{R}_0 u - 1)}{\mathcal{R}_0^2 u^2 F(u)^2} \left[Y_1(u) - \frac{h_\theta(u)}{\mathcal{R}_0 u} \right].$$

Then the two outer coefficients are the nested one-dimensional integrals

$$\boxed{Y_1(x) = \int_1^x q_1(u) du, \quad Y_2(x) = \int_1^x q_2(u) du.} \quad (5)$$

These are exactly equivalent to $Y'_i = q_i$, $Y_i(1) = 0$. Numerically, an ODE integrator is simply an adaptive quadrature engine for these cumulative integrals. The apparent singularity at $u = 1$ is removable. Near $u = x_f$ the raw second integral is badly conditioned; the production calculation therefore subtracts its known singular part, integrates the finite remainder, and adds back the analytic primitive, as detailed in Section 8. For $0 < y < F(x_*)$ let $X_0(y) \in (x_f, x_*)$ be the descending-branch solution of $F(X_0) = y$, and set

$$X_1(y) = -\frac{Y_1(X_0)}{F'(X_0)},$$

$$X_2(y) = -\frac{\frac{1}{2}F''(X_0)X_1^2 + Y_1'(X_0)X_1 + Y_2(X_0)}{F'(X_0)}.$$

Define the local constants

$$\begin{aligned} \delta &= 1 - \mathcal{R}_0 x_f, & h_f &= h_\theta(x_f), & h'_f &= x_f^{\theta-1} \{ \theta - (\theta + 1)x_f \}, \\ a &= \frac{h_f}{\delta}, & b &= \frac{\mathcal{R}_0 x_f}{\delta}, & c &= \delta h'_f + \mathcal{R}_0 h_f, \\ \kappa &= \frac{h_f c}{2\delta^3}, & f_1 &= \frac{\delta}{\mathcal{R}_0 x_f}, & f_2 &= -\frac{1}{\mathcal{R}_0 x_f^2}, \\ \alpha &= \frac{h_f}{\mathcal{R}_0 x_f}, & \beta &= \alpha \log \frac{f_1}{C_\theta}, & p_0 &= \frac{2\delta x_f h'_f - (1 + 2\delta)h_f}{2\mathcal{R}_0 \delta x_f^2}, \\ A_{-1} &= \frac{h_f/x_f - c}{\delta^2}, & B_0 &= \beta - \alpha, & B_1 &= -\frac{h_f}{2\mathcal{R}_0 \delta x_f^2}. \end{aligned}$$

For $s > 0$, subtract the explicitly known singular part

$$q_{2,\text{sing}}(s) = -\frac{a\alpha \log s + aB_0}{s^2} + \frac{A_{-1}\alpha \log s + A_{-1}B_0 - aB_1}{s}, \quad (6)$$

and define

$$\begin{aligned} q_{2,\text{reg}}(u) &= q_2(u) - q_{2,\text{sing}}(u - x_f), \\ P_2(s) &= \frac{a\alpha \log s + a\beta}{s} + \frac{A_{-1}\alpha}{2} \log^2 s + (A_{-1}B_0 - aB_1) \log s, \\ K_2 &= -P_2(1 - x_f) - \int_{x_f}^1 q_{2,\text{reg}}(u) du, \\ N_{20} &= K_2 + \frac{\alpha\beta f_2}{f_1^2} - \frac{(\alpha + \beta)p_0}{f_1} + \frac{f_2\beta^2}{2f_1^2}. \end{aligned} \quad (7)$$

The production formula for the second-order constant is therefore

$$\boxed{D_\theta = -\frac{N_{20}}{f_1} - \kappa \frac{\beta^2}{\alpha^2}.} \quad (8)$$

The integrand $q_{2,\text{reg}}(u) = O(\log(u - x_f))$ is integrable at $u = x_f$, so (8) requires one finite regularized quadrature and no small- y extrapolation. For general θ this is the explicit semi-analytical expression; an elementary closed form is not available. The implementation stops a short distance from each endpoint and supplies the omitted pieces from their local log-power expansions. It also normalizes the numerically integrated Y_1 so that its finite part is exactly $\beta = \alpha \log(f_1/C_\theta)$. These are quadrature-stabilization steps only: the endpoint distances, fitted local coefficients, and normalization shift do not alter the finite part D_θ and are not extra model parameters. More precisely, the local outer calculation in Section 8 gives

$$X_2(y) = \kappa \log^2\left(\frac{C_\theta}{y}\right) + D_\theta + o(1), \quad y \downarrow 0,$$

with D_θ equal to (8). In the overlap with the corner region, this outer expansion is compared with the corner solution containing an undetermined integration constant. Equality of the constant terms fixes that corner constant to D_θ , which is why the same quantity appears in the entry-point formulas below.

1.3 First- and second-order entry points

Only differences of the action are needed. Define them directly by

$$\mathcal{A}_\theta(x) - \mathcal{A}_\theta(z) = \int_z^x \frac{1 - \mathcal{R}_0 u}{(1 - u)u^\theta} du, \quad x_f \leq z, x \leq x_*, \quad (9)$$

and put

$$\begin{aligned} L_{\text{BL}} &= \log(C_\theta/y_{\text{BL}}), & Q &= \frac{\mathcal{R}_0 a - bc}{a\delta^2}, \\ M_1 &= \rho L_{\text{BL}} + \frac{b}{a} y_{\text{BL}}, \\ M_2 &= M_1 + \rho^2 \frac{D_\theta}{a} + \rho y_{\text{BL}} Q (L_{\text{BL}} + 1) + \frac{bQ}{2a} y_{\text{BL}}^2. \end{aligned}$$

The direct corner expansions for the entry point are, explicitly,

$$\boxed{x_{\text{in}}^{(1), \text{dir}} = x_f + \rho a L_{\text{BL}} + b y_{\text{BL}},} \quad (10)$$

and

$$\boxed{\begin{aligned} x_{\text{in}}^{(2), \text{dir}} &= x_f + \rho a L_{\text{BL}} + b y_{\text{BL}} + \rho^2 \{\kappa L_{\text{BL}}^2 + D_\theta\} \\ &\quad + \frac{\rho y_{\text{BL}}}{\delta^2} \{\mathcal{R}_0 a (L_{\text{BL}} + 1) - bc\} + \frac{b \mathcal{R}_0}{2\delta^2} y_{\text{BL}}^2. \end{aligned}} \quad (11)$$

These two formulas display the asymptotic corrections transparently. For production calculations we recommend the action-resummed versions: they keep the same first- or second-order matching information but avoid unnecessarily Taylor-expanding the nonlinear action. The recommended first- and second-order entry estimates are therefore the unique admissible roots

$$\boxed{\mathcal{A}_\theta(x_{\text{in}}^{(j)}) - \mathcal{A}_\theta(x_f) = M_j, \quad x_f < x_{\text{in}}^{(j)} < x_*, \quad j = 1, 2.} \quad (12)$$

They exist precisely when

$$0 < M_j < \Delta A_f, \quad \Delta A_f = \mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_f). \quad (13)$$

Failure of (13) is a validity diagnostic, not a license to extrapolate the scalar equation beyond its matching interval.

1.4 Burnout and persistence probabilities

For either $x_{\text{in}} = x_{\text{in}}^{(1)}$ or $x_{\text{in}}^{(2)}$, define the exact one-dimensional Kendall integral

$$\mathcal{I}(x_{\text{in}}) = \int_{x_{\text{in}}}^1 \frac{1}{\rho(1-X)X^\theta} \exp\left\{\frac{\mathcal{A}_\theta(X) - \mathcal{A}_\theta(x_{\text{in}})}{\rho}\right\} dX. \quad (14)$$

The production calculation evaluates the same integral in the overflow-safe form

$$\begin{aligned} \Lambda(x_{\text{in}}) &= \frac{\mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_{\text{in}})}{\rho}, \\ J(x_{\text{in}}) &= \int_{x_{\text{in}}}^1 \frac{1}{\rho(1-X)X^\theta} \exp\left\{\frac{\mathcal{A}_\theta(X) - \mathcal{A}_\theta(x_*)}{\rho}\right\} dX, \\ \log \mathcal{I} &= \Lambda + \log J. \end{aligned} \quad (15)$$

Thus J is only a rescaled numerical integral, not an additional approximation or an unreported model quantity. With $m = Ky_{\text{BL}}$ infectives at entry, the conditional probabilities are

$$\boxed{P_{\text{burn|est}} \simeq \left(\frac{\mathcal{I}}{1 + \mathcal{I}}\right)^{Ky_{\text{BL}}}, \quad P_{\text{persist|est}} \simeq 1 - P_{\text{burn|est}}.} \quad (16)$$

Numerically these are evaluated without forming a potentially enormous $\mathcal{I}$:

$$B = m \log(1 + e^{-\log \mathcal{I}}), \quad P_{\text{persist|est}} = -\text{expm1}(-B), \quad P_{\text{burn|est}} = e^{-B}. \quad (17)$$

For one initially infectious individual, the early establishment probability is $p_{\text{est}} \simeq 1 - 1/\mathcal{R}_0$, so the unconditional first-trough persistence probability is

$$P_{\text{persist,uncond}} \simeq \left(1 - \frac{1}{\mathcal{R}_0}\right) P_{\text{persist|est}}. \quad (18)$$

For I_0 independent initial lineages, replace the leading factor by $1 - \mathcal{R}_0^{-I_0}$.

The optional Laplace closed form uses

$$\Lambda_j = \frac{\Delta A_f - M_j}{\rho}, \quad \mathcal{I}_{L,j} = \sqrt{\frac{2\pi}{\mathcal{R}_0 y^*}} e^{\Lambda_j}, \quad (19)$$

followed by (16) with $\mathcal{I}_{L,j}$ in place of $\mathcal{I}$. The further simplification $-\log P_{\text{burn|est}} \sim Ky_{\text{BL}}/\mathcal{I}_{L,j}$ additionally requires $\mathcal{I}_{L,j} \gg 1$.

Boundary-layer-independent leading and next order. Strict-overlap matching also supplies a closed form in which the arbitrary matching height cancels. Combining

$$\mathcal{A}_\theta(x_{\text{in}}) - \mathcal{A}_\theta(x_f) \sim \rho \log\left(\frac{C_\theta}{y_{\text{BL}}}\right), \quad y_{\text{min}} = y_{\text{BL}} \exp\left\{-\frac{\mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_{\text{in}})}{\rho}\right\} \quad (20)$$

gives

$$\boxed{y_{\text{min}}^{\text{BI}} = C_\theta \exp\left(-\frac{\Delta A_f}{\rho}\right).} \quad (21)$$

The generalized trough/Laplace burnout approximation is therefore

$$\boxed{B_{\text{BI}} = KC_\theta \sqrt{\frac{\mathcal{R}_0 y^*}{2\pi}} \exp\left(-\frac{\Delta A_f}{\rho}\right), \quad P_{\text{persist|est}}^{\text{BI}} = -\text{expm1}(-B_{\text{BI}}).} \quad (22)$$

The unconditional probability multiplies this by $1 - 1/\mathcal{R}_0$ for one initial infective, or by $1 - \mathcal{R}_0^{-I_0}$ for I_0 independent initial infectives. This is genuinely boundary-layer-independent at leading order and remains defined when the finite-boundary construction reports no entry. It is an additional leading-order prediction, not a presumed replacement for either finite-boundary route.

At the next order, impose the stronger formal overlap

$$\frac{1}{K} \ll y_{\text{BL}} \ll \rho^2. \quad (23)$$

Then the ρy_{BL} and y_{BL}^2 terms in (11) are beyond the retained $O(\rho^2)$ action accuracy, and (12) gives

$$\mathcal{A}_\theta(x_{\text{in}}) - \mathcal{A}_\theta(x_f) = \rho \log(C_\theta/y_{\text{BL}}) + \rho^2 D_\theta/a + o(\rho^2).$$

Consequently the cancellation is explicit:

$$\begin{aligned} y_{\text{min}}^{\text{BI},2} &= y_{\text{BL}} \exp \left[-\frac{\Delta A_f - \rho \log(C_\theta/y_{\text{BL}}) - \rho^2 D_\theta/a + o(\rho^2)}{\rho} \right] \\ &= \underbrace{y_{\text{BL}} \frac{C_\theta}{y_{\text{BL}}}}_{C_\theta} \exp \left[-\frac{\Delta A_f}{\rho} + \rho \frac{D_\theta}{a} + o(\rho) \right] \\ &\sim C_\theta \exp \left[-\frac{\Delta A_f}{\rho} + \rho \frac{D_\theta}{a} \right]. \end{aligned} \quad (24)$$

There is no y_{BL} in the final expression.

Writing $h_* = h_\theta(1/\mathcal{R}_0)$ and using primes for x derivatives at the saddle, the next Laplace coefficient is

$$c_L = \frac{h_*}{\mathcal{R}_0} \left\{ \frac{1}{12} \left(\frac{h'_*}{h_*} \right)^2 - \frac{1}{8} \frac{h''_*}{h_*} \right\}. \quad (25)$$

Thus the next-order boundary-independent closed form is

$$\boxed{B_{\text{BI}}^{\text{next}} = K C_\theta \sqrt{\frac{\mathcal{R}_0 y^*}{2\pi}} \exp \left[-\frac{\Delta A_f}{\rho} + \rho \left(\frac{D_\theta}{a} - c_L \right) \right], \quad P_{\text{persist}|\text{est}} = 1 - e^{-B_{\text{BI}}^{\text{next}}}.} \quad (26)$$

This is a next-order boundary-independent approximation, not a uniform second-order theory. It retains the fixed- $\mathcal{R}_0 > 1$, small- ρ , large- K assumptions and remains nonuniform as $\mathcal{R}_0 \rightarrow 1^+$.

Which route requires x_{in} ? There are two distinct computational pipelines.

1. **Exact Kendall (recommended high-accuracy route):** compute C_θ, D_θ and M_j ; solve $\mathcal{A}_\theta(x_{\text{in}}^{(j)}) - \mathcal{A}_\theta(x_f) = M_j$ for $x_{\text{in}}^{(j)}$; then use that value as the lower limit in (14). Thus this route genuinely requires x_{in} .
2. **Laplace closed form:** compute the action directly as

$$\Lambda_j = \frac{\Delta A_f - M_j}{\rho}.$$

Then use (19) and (16). No root solve for x_{in} and no separate evaluation of $\mathcal{A}_\theta(x_{\text{in}})$ are required, because the matching equation has already supplied their action difference.

1.5 Conditions: what is required, and why

The conditions are best treated as diagnostics rather than as a single vague statement that every parameter must be “small” or “large.”

Common to both orders.

1. $\rho \ll 1$ with fixed $\mathcal{R}_0 > 1$, and in practice $(\mathcal{R}_0 - 1)/\rho \gg 1$. This separates the fast epidemic wave from slow susceptible replenishment and avoids the nonuniform near-threshold limit.
2. $1/K \ll y_{\text{BL}} \ll \rho$ (hence $Ky_{\text{BL}} \gg 1$ and $Ky^* \gg 1$ for the standard choice). The first inequality makes deterministic entry concentrated enough to match to a branching process; the second makes infection feedback on the susceptible path negligible in the Kendall layer.
3. The deterministic descending trajectory must actually cross $y = y_{\text{BL}}$, and (13) must hold. Otherwise there is no entry point of the type assumed by the calculation.
4. The fixed- $\mathcal{R}_0$ expansion must not be pushed into the joint high- $\mathcal{R}_0$ regime. For $0 \leq \theta < 1$, a useful warning test is $x_f \gg \rho^{1/(1-\theta)}$; if these scales compete, replenishment changes the final-size corner itself. The endpoint $\theta = 1$ has a different distinguished balance and must not be judged by this formula.

First-order matching. In addition to the common conditions, the corner displacement must be small:

$$\rho|L_{\text{BL}}| + y_{\text{BL}} \ll x_* - x_f, \quad 0 < M_1 < \Delta A_f. \quad (27)$$

This is the operational requirement that the logarithmic switchback and the finite-height correction remain local to the final-size corner. It is the appropriate accuracy level corresponding to the refined formula in the original burnout paper when $\theta = 0$.

Second-order matching. All first-order conditions remain necessary. In addition, the nominal second-order correction

$$\Delta M_2 = M_2 - M_1 = \rho^2 \frac{D_\theta}{a} + \rho y_{\text{BL}} Q(L_{\text{BL}} + 1) + \frac{bQ}{2a} y_{\text{BL}}^2 \quad (28)$$

must be hierarchically smaller than the leading displacement, for example

$$|\Delta M_2| \ll |M_1|, \quad 0 < M_2 < \Delta A_f. \quad (29)$$

This check explains why a formal second-order curve can be worse than the first-order curve at moderate ρ : the extra terms are then no longer small corrections and may over-correct. Second order improves deterministic matching only; it cannot cure finite- K stochastic-entry error or a poor Laplace approximation.

Laplace approximation only. The exact Kendall integral (14) needs no saddle-point assumption. Replacing it by (19) additionally requires $\Lambda_j \gg 1$, so that the integrand is sharply localized near $x = x_*$. When $\Lambda_j = O(1)$, retain the exact integral even if the entry estimate is second order.

2 The problem: from a deterministic epidemic tail to stochastic burnout

We consider the dimensionless SIR-type dynamics in a single patch

$$\frac{dx}{d\sigma} = \rho h_\theta(x) - \mathcal{R}_0 x y, \quad \frac{dy}{d\sigma} = (\mathcal{R}_0 x - 1)y, \quad (30)$$

where

$$h_\theta(x) = (1 - x)x^\theta, \quad 0 \leq \theta \leq 1. \quad (31)$$

Here x and y are the susceptible and infectious fractions, respectively, $\mathcal{R}_0 > 1$ is the basic reproduction number, and ρ controls the slow time scale of susceptible replenishment. The case $\theta = 0$ gives the standard linear replenishment model, while $\theta = 1$ gives logistic replenishment; intermediate values interpolate continuously between the two.

At low susceptible density,

$$h_\theta(x) \sim x^\theta, \quad x \rightarrow 0,$$

so θ may be interpreted as the low-density elasticity of recruitment with respect to the current susceptible fraction. The range $0 \leq \theta \leq 1$ therefore includes both the Parsons model and the logistic limit.

Define the epidemic growth threshold

$$x_* = \frac{1}{\mathcal{R}_0}. \quad (32)$$

The endemic equilibrium satisfies $x = x_*$ and

$$y^* = \rho h_\theta(x_*) = \rho(\mathcal{R}_0 - 1)\mathcal{R}_0^{-(1+\theta)}. \quad (33)$$

Let K denote the characteristic host population size of one patch. After the first epidemic wave, the number of infectious individuals falls from a macroscopic level into a stochastic branching regime. To match the deterministic epidemic tail to the low-count stochastic process, we use the practical matching height

$$y_{\text{BL}} = \sqrt{\frac{y^*}{K}}. \quad (34)$$

When $Ky^* \gg 1$,

$$\frac{1}{K} \ll y_{\text{BL}} \ll y^* \sim \rho,$$

so y_{BL} lies in an overlap region between the one-infectious-individual scale and the deterministic endemic prevalence.

We define x_{in} as the susceptible fraction at the first downward crossing of $y = y_{\text{BL}}$ after the first epidemic peak. Our goal is to construct the map

$$(\mathcal{R}_0, \rho, \theta, K) \longrightarrow x_{\text{in}} \longrightarrow P_1,$$

without integrating the full two-dimensional system (30) for every parameter set.

3 The Parsons framework and the logic of the present extension

Parsons and Earn constructed matched asymptotic approximations for the standard SIR model with vital dynamics, including the epidemic outer solution, the left corner, the x -axis boundary layer, and an additional boundary layer required in a joint high- $\mathcal{R}_0$ limit[1]. Parsons et al. then connected this phase-plane construction to a time-inhomogeneous branching process to obtain the probability of burnout during the first epidemic trough[2].

The present work uses the same singular-perturbation architecture but relies only on two structural facts:

1. when $\rho = 0$, the geometry of the first epidemic wave is independent of h_θ ;
2. when y is small, susceptible replenishment enters the slow basin dynamics through $h_\theta(x)$.

Hence the same outer/corner hierarchy can be extended from standard linear replenishment to the whole family $h_\theta(x) = (1-x)x^\theta$.

The main asymptotic regime considered here is

$$\rho \rightarrow 0, \quad \mathcal{R}_0 > 1 \text{ fixed}, \quad K \text{ large enough that } 1/K \ll y_{\text{BL}} \ll \rho. \quad (35)$$

The joint high- $\mathcal{R}_0$ limit and finite- K stochastic entry are treated separately as boundaries of validity.

4 Leading order: universal epidemic geometry

Set $\rho = 0$. Equation (30) gives

$$\frac{dy}{dx} = \frac{1}{\mathcal{R}_0 x} - 1.$$

Starting from a naive invasion $(x, y) = (1, 0^+)$, the first-wave outer trajectory is

$$F(x) = 1 - x + x_* \log x, \quad y = F(x). \quad (36)$$

The epidemic peak occurs at $x = x_*$. The zero-demography final susceptible fraction x_f is the smaller root of $F(x_f) = 0$:

$$x_f = -\frac{1}{\mathcal{R}_0} W_0(-\mathcal{R}_0 e^{-\mathcal{R}_0}), \quad (37)$$

where W_0 is the principal branch of the Lambert W function.

The point x_f is the anchor of the corner matching, but it cannot be used directly as the finite- ρ value of x_{in} . The reason is that the later burnout probability depends on an action scaled by $1/\rho$; even if $x_{\text{in}} - x_f \rightarrow 0$, that difference may still be exponentially amplified. A systematic matched correction to x_{in} is therefore required.

5 Exact phase-plane equation and the direct perturbation hierarchy

From (30), the exact phase-plane equation is

$$\frac{dy}{dx} = \frac{(\mathcal{R}_0 x - 1)y}{\rho h_\theta(x) - \mathcal{R}_0 x y}. \quad (38)$$

In the first-wave outer region, write the direct expansion

$$y = F(x) + \rho Y_1(x) + \rho^2 Y_2(x) + O(\rho^3). \quad (39)$$

Substituting (39) into (38) and equating successive powers of ρ gives

$$Y_1'(x) = -\frac{h_\theta(x)(\mathcal{R}_0 x - 1)}{\mathcal{R}_0^2 x^2 F(x)}, \quad Y_1(1) = 0, \quad (40)$$

and

$$Y_2'(x) = \frac{h_\theta(x)(\mathcal{R}_0 x - 1)}{\mathcal{R}_0^2 x^2 F(x)^2} \left[Y_1(x) - \frac{h_\theta(x)}{\mathcal{R}_0 x} \right], \quad Y_2(1) = 0. \quad (41)$$

The apparent singularities at $x = 1$ are removable. Up to second order, the outer correction is therefore determined by two one-dimensional coefficient equations.

An equivalent bookkeeping device uses the first integral of the $\rho = 0$ SIR system,

$$H = x + y - \frac{1}{\mathcal{R}_0} \log x.$$

Along the full model,

$$\frac{dH}{d\sigma} = \rho h_\theta(x) \left(1 - \frac{1}{\mathcal{R}_0 x}\right).$$

If one writes $H = 1 + \rho H_1 + \rho^2 H_2 + \dots$, then $H_1 = Y_1$ and $H_2 = Y_2$. Thus the conserved-quantity formulation is simply an equivalent organization of the same direct phase-plane hierarchy, not a separate approximation route.

6 First-order matching: the logarithmic singularity and C_θ

6.1 Relation to the Parsons notation

Parsons and Earn use two visually similar small parameters. Their first definition, introduced in the time-parametrized SIR equations, is

$$\varepsilon = \frac{\mu}{\gamma + \mu},$$

the infectious period divided by the host lifetime. In the standard recruitment case $\theta = 0$, their equation is

$$\frac{dx}{d\sigma} = \varepsilon(1 - x) - \mathcal{R}_0 xy,$$

whereas ours is $dx/d\sigma = \rho(1 - x) - \mathcal{R}_0 xy$. Therefore the exact dictionary is

$$\boxed{\varepsilon = \rho.} \tag{42}$$

They then introduce the second glyph

$$\boxed{\epsilon = \frac{\varepsilon}{\mathcal{R}_0} = \frac{\rho}{\mathcal{R}_0},} \tag{43}$$

because it simplifies the phase-plane equations. Thus ε and ϵ are distinct, with $\varepsilon = \mathcal{R}_0 \epsilon$; neither is an additional biological parameter. This note expands directly in ρ ,

$$y = F + \rho Y_1 + \rho^2 Y_2 + \dots,$$

while Parsons and Earn organize the corresponding phase-plane expansion in ϵ . Since $\mathcal{R}_0$ is fixed, $\epsilon = O(\rho)$ and the asymptotic orders agree. To translate their first-order coefficient to ours, set

$$G_\theta = \mathcal{R}_0 Y_1. \tag{44}$$

Then

$$G'_\theta(x) = \frac{(x_* - x)h_\theta(x)}{x^2 F(x)}. \tag{45}$$

Define

$$\tilde{a}_\theta = \frac{h_\theta(x_f)}{x_f} = (1 - x_f)x_f^{\theta-1}. \tag{46}$$

Because

$$F'(x_f) = \frac{x_* - x_f}{x_f},$$

we have, as $x \rightarrow x_f^+$,

$$G'_\theta(x) \sim \frac{\tilde{a}_\theta}{x - x_f}.$$

Thus the first-order outer correction necessarily contains a logarithmic singularity.

6.2 The regularized finite part

Define the regularized integral

$$\mathcal{J}_\theta = \int_{x_f}^1 \left[\frac{(x_* - u)h_\theta(u)}{u^2 F(u)} - \frac{\tilde{a}_\theta}{u - x_f} \right] du. \quad (47)$$

The two singular terms cancel at $u = x_f$, so (47) is a smooth one-dimensional integral. Its finite part determines the matching amplitude as follows. Since $G_\theta(1) = 0$, integration of (45) gives

$$G_\theta(x) = - \int_x^1 \frac{(x_* - u)h_\theta(u)}{u^2 F(u)} du. \quad (48)$$

Adding and subtracting the singular term used in (47), and then taking $x \rightarrow x_f^+$, yields

$$G_\theta(x) = \tilde{a}_\theta \log \frac{x - x_f}{1 - x_f} - \mathcal{J}_\theta + o(1). \quad (49)$$

To convert this result from the direct representation $y = y(x; \rho)$ to the inverse descending branch $x = X(y; \rho)$, introduce

$$X(y; \rho) = X_0(y) + \rho X_1(y) + \rho^2 X_2(y) + O(\rho^3), \quad F(X_0(y)) = y. \quad (50)$$

Substitute (50) into the direct outer expansion (39). Taylor expansion at fixed y gives, through first order,

$$y = F(X_0) + \rho [F'(X_0)X_1 + Y_1(X_0)] + O(\rho^2).$$

The leading term is $F(X_0) = y$ by definition, so the coefficient of ρ must vanish. Hence

$$X_1(y) = - \frac{Y_1(X_0(y))}{F'(X_0(y))} = - \frac{G_\theta(X_0(y))}{\mathcal{R}_0 F'(X_0(y))}. \quad (51)$$

This is the usual first-order inverse-function correction; it requires $F'(X_0) \neq 0$, which holds on the descending branch sufficiently close to x_f because $F'(x_f) > 0$. The relation needed below is obtained by Taylor-expanding the identity $F(X_0(y)) = y$ about x_f :

$$y = F(x_f) + F'(x_f)(X_0(y) - x_f) + O((X_0(y) - x_f)^2).$$

Since $F(x_f) = 0$ and $F'(x_f) \neq 0$, inversion gives

$$X_0(y) - x_f = \frac{y}{F'(x_f)} + O(y^2), \quad F'(x_f) = \frac{x_* - x_f}{x_f}. \quad (52)$$

Thus, in (49), setting $x = X_0(y)$ and using $X_0(y) - x_f = y/F'(x_f) + O(y^2)$ gives explicitly

$$G_\theta(X_0(y)) = \tilde{a}_\theta \log \frac{y}{(1 - x_f)F'(x_f)} - \mathcal{J}_\theta + o(1). \quad (53)$$

Substituting (53) into (51) first gives

$$X_1(y) = - \frac{1}{\mathcal{R}_0 F'(x_f)} \left[\tilde{a}_\theta \log \frac{y}{(1 - x_f)F'(x_f)} - \mathcal{J}_\theta \right] + o(1),$$

where $F'(X_0(y)) = F'(x_f) + O(y)$ has also been used. Expanding the minus sign and factoring out $\tilde{a}_\theta$ gives

$$X_1(y) = \frac{\tilde{a}_\theta}{\mathcal{R}_0 F'(x_f)} \left[\log \frac{(1 - x_f)F'(x_f)}{y} + \frac{\mathcal{J}_\theta}{\tilde{a}_\theta} \right] + o(1).$$

Finally, using $\log A + B = \log(Ae^B)$ gives

$$X_1(y) = \frac{\tilde{a}_\theta}{\mathcal{R}_0 F'(x_f)} \log \left[\frac{(1-x_f)F'(x_f) \exp(\mathcal{J}_\theta/\tilde{a}_\theta)}{y} \right] + o(1). \quad (54)$$

The numerator inside the logarithm in (54) is independent of y and is positive. We define this scale to be the first-order matching constant

$$C_\theta = (1-x_f)F'(x_f) \exp\left(\frac{\mathcal{J}_\theta}{\tilde{a}_\theta}\right) = \frac{(1-x_f)(x_*-x_f)}{x_f} \exp\left(\frac{\mathcal{J}_\theta}{\tilde{a}_\theta}\right). \quad (55)$$

Thus C_θ is the positive scale selected by the finite part of the outer solution. Moreover,

$$\frac{\tilde{a}_\theta}{\mathcal{R}_0 F'(x_f)} = \frac{h_\theta(x_f)}{1 - \mathcal{R}_0 x_f} = a.$$

Consequently, (54) takes the compact asymptotic form

$$X_1(y) = a \log \frac{C_\theta}{y} + o(1), \quad y \downarrow 0. \quad (56)$$

The constant C_θ depends only on $(\mathcal{R}_0, \theta)$ and is independent of (ρ, K) .

When $\theta = 0$, (47) reduces to the regularized integral in the refined Parsons et al. burnout estimate, and (55) reduces to its first-order prefactor[2]. Hence C_θ is a direct generalization of the standard burnout matching constant.

7 Inverse outer expansion and corner scaling

The inverse descending-branch expansion and its first-order coefficient were introduced in (50) and (51). To derive the second-order coefficient of the inverse trajectory, substitute $X = X_0 + \rho X_1 + \rho^2 X_2 + O(\rho^3)$ into the identity

$$y = F(X) + \rho Y_1(X) + \rho^2 Y_2(X) + O(\rho^3).$$

Taylor expansion about X_0 gives

$$\begin{aligned} F(X) &= F(X_0) + \rho F'(X_0)X_1 \\ &\quad + \rho^2 \left[F'(X_0)X_2 + \frac{1}{2}F''(X_0)X_1^2 \right] + O(\rho^3), \\ \rho Y_1(X) &= \rho Y_1(X_0) + \rho^2 Y_1'(X_0)X_1 + O(\rho^3), \\ \rho^2 Y_2(X) &= \rho^2 Y_2(X_0) + O(\rho^3). \end{aligned}$$

Equating coefficients of like powers of ρ gives $F(X_0) = y$ at $O(1)$ and (51) at $O(\rho)$. Equating the coefficients of ρ^2 gives

$$F'(X_0)X_2 + \frac{1}{2}F''(X_0)X_1^2 + Y_1'(X_0)X_1 + Y_2(X_0) = 0.$$

Solving this equation for X_2 gives

$$X_2(y) = -\frac{\frac{1}{2}F''(X_0)X_1^2 + Y_1'(X_0)X_1 + Y_2(X_0)}{F'(X_0)}. \quad (57)$$

The outer expansion is no longer valid near $y = 0$ because the first-order term $\rho X_1(y)$ is no longer small relative to the leading displacement $X_0(y) - x_f$. Equations (52) and (56) give

$$X_0(y) - x_f \sim \frac{y}{F'(x_f)}, \quad \rho X_1(y) \sim \rho a \log \frac{C_\theta}{y}.$$

The outer inverse expansion requires the correction ρX_1 to be small relative to the leading displacement $X_0 - x_f$. Their magnitude ratio behaves as

$$\frac{\rho|X_1(y)|}{|X_0(y) - x_f|} \sim \frac{\rho a F'(x_f)}{y} \left| \log \frac{C_\theta}{y} \right|,$$

which is not small when y reaches the ρ -scale and diverges as $y \downarrow 0$. Thus the actual trajectory remains finite, but the outer ordering is no longer uniform near x_f . To resolve this region, introduce the corner variables

$$y = \rho v, \quad x = x_f + \rho u_1(v) + \rho^2 u_2(v) + \dots \quad (58)$$

Define

$$\delta = 1 - \mathcal{R}_0 x_f, \quad h_f = h_\theta(x_f), \quad h'_f = h'_\theta(x_f), \quad (59)$$

and

$$a = \frac{h_f}{\delta}, \quad b = \frac{\mathcal{R}_0 x_f}{\delta}, \quad c = \delta h'_f + \mathcal{R}_0 h_f, \quad \kappa = \frac{h_f c}{2\delta^3}. \quad (60)$$

For $h_\theta(x) = (1 - x)x^\theta$,

$$h'_\theta(x) = x^{\theta-1} \{\theta - (\theta + 1)x\}.$$

Let

$$L = \log \frac{C_\theta}{y}. \quad (61)$$

To derive the first corner coefficient, invert the exact phase-plane equation (38):

$$\frac{dx}{dy} = \frac{\rho h_\theta(x) - \mathcal{R}_0 x y}{(\mathcal{R}_0 x - 1)y}. \quad (62)$$

Under (58), $y = \rho v$ and $x = x_f + \rho u_1(v) + \rho^2 u_2(v) + O(\rho^3)$. Holding ρ fixed and differentiating both variables with respect to v gives

$$\frac{dy}{dv} = \rho, \quad \frac{dx}{dv} = \rho u'_1(v) + \rho^2 u'_2(v) + O(\rho^3).$$

The chain rule therefore gives

$$\frac{dx}{dy} = \frac{dx/dv}{dy/dv} = u'_1(v) + \rho u'_2(v) + O(\rho^2) = u'_1(v) + O(\rho).$$

The remaining factors in the inverse phase-plane equation satisfy

$$h_\theta(x) = h_f + O(\rho), \quad \mathcal{R}_0 x - 1 = -\delta + O(\rho).$$

Substitution therefore gives, at leading order,

$$u'_1(v) = \frac{h_f - \mathcal{R}_0 x_f v}{-\delta v} = b - \frac{a}{v}.$$

Integration yields

$$u_1(v) = bv - a \log v + A(\rho),$$

where the local corner equation does not determine the integration constant $A(\rho)$. The outer expansion and the corner expansion are two approximations to the same trajectory. In their overlap, both approximations are valid and must have the same asymptotic form. The outer expansion already contains the constant C_θ , whereas the corner expansion contains the undetermined $A(\rho)$; matching the two representations determines $A(\rho)$ in terms of C_θ .

In the overlap $\rho \ll y \ll 1$ (equivalently, $v = y/\rho \rightarrow \infty$ while $y \rightarrow 0$), use $y = \rho v$, $1/F'(x_f) = b$, and (56) to express the outer expansion in corner variables:

$$x - x_f \sim \frac{y}{F'(x_f)} + \rho a \log \frac{C_\theta}{y} = \rho \left[bv - a \log v + a \log \frac{C_\theta}{\rho} \right].$$

Comparison with $x - x_f = \rho u_1(v) + \dots$ gives $A(\rho) = a \log(C_\theta/\rho)$. Consequently,

$$u_1(v) = bv - a \log v + a \log \frac{C_\theta}{\rho} = a \log \frac{C_\theta}{\rho v} + bv = aL + bv. \quad (63)$$

The term ρaL is the logarithmic switchback, while $\rho bv = by$ is the finite-height correction. Equation (63) is therefore the corner representation of the refined first-order outer result, with C_θ supplied by outer matching.

8 Second-order matching: D_θ and the next switchback

The second corner coefficient follows by retaining one more order in the inverse phase-plane equation (62). Denote the numerator and denominator on its right-hand side by

$$N(x, y; \rho) = \rho h_\theta(x) - \mathcal{R}_0 x y, \quad D(x, y) = (\mathcal{R}_0 x - 1)y,$$

so that $dx/dy = N/D$. Substitute $x = x_f + \rho u_1 + \rho^2 u_2 + \dots$ and $y = \rho v$. First,

$$h_\theta(x) = h_f + \rho h'_f u_1 + O(\rho^2),$$

and therefore the numerator is

$$\begin{aligned} N(x, y; \rho) &= \rho [h_f + \rho h'_f u_1 + O(\rho^2)] - \mathcal{R}_0 [x_f + \rho u_1 + O(\rho^2)] \rho v \\ &= \rho(h_f - \mathcal{R}_0 x_f v) + \rho^2 u_1(h'_f - \mathcal{R}_0 v) + O(\rho^3). \end{aligned}$$

Similarly,

$$\begin{aligned} D(x, y) &= [-\delta + \rho \mathcal{R}_0 u_1 + O(\rho^2)] \rho v \\ &= -\rho \delta v + \rho^2 \mathcal{R}_0 u_1 v + O(\rho^3). \end{aligned}$$

Write these results as

$$N = \rho N_0 + \rho^2 N_1 + O(\rho^3), \quad D = \rho D_0 + \rho^2 D_1 + O(\rho^3),$$

where

$$N_0 = h_f - \mathcal{R}_0 x_f v, \quad N_1 = u_1(h'_f - \mathcal{R}_0 v), \quad D_0 = -\delta v, \quad D_1 = \mathcal{R}_0 u_1 v.$$

The quotient expansion is

$$\frac{N}{D} = \frac{N_0}{D_0} + \rho \left(\frac{N_1}{D_0} - \frac{N_0 D_1}{D_0^2} \right) + O(\rho^2).$$

Its leading term equals $u'_1 = N_0/D_0$. Comparing its $O(\rho)$ coefficient with $dx/dy = u'_1 + \rho u'_2 + O(\rho^2)$ gives

$$\begin{aligned} u'_2 &= \frac{N_1}{D_0} - \frac{N_0 D_1}{D_0^2} \\ &= -\frac{u_1(h'_f - \mathcal{R}_0 v)}{\delta v} - \frac{\mathcal{R}_0 u_1(h_f - \mathcal{R}_0 x_f v)}{\delta^2 v}. \end{aligned}$$

Using $\delta + \mathcal{R}_0 x_f = 1$ and $c = \delta h'_f + \mathcal{R}_0 h_f$ reduces this expression to

$$u'_2(v) = \frac{u_1(v)}{\delta^2} \left(\mathcal{R}_0 - \frac{c}{v} \right), \quad c = \delta h'_f + \mathcal{R}_0 h_f. \quad (64)$$

Substitute $u_1 = aL + bv$ and use $dL/dv = -1/v$. Since $ac/\delta^2 = 2\kappa$, (64) becomes

$$u'_2 = -\frac{2\kappa L}{v} + \frac{\mathcal{R}_0 a}{\delta^2} L - \frac{bc}{\delta^2} + \frac{b\mathcal{R}_0}{\delta^2} v.$$

Each term now integrates directly:

$$\int -\frac{2\kappa L}{v} dv = \kappa L^2, \quad \int L dv = (L+1)v.$$

Therefore

$$u_2(v) = \kappa L^2 + B_2(\rho) + \frac{\mathcal{R}_0 a}{\delta^2} (L+1)v - \frac{bc}{\delta^2} v + \frac{b\mathcal{R}_0}{2\delta^2} v^2. \quad (65)$$

The local equation determines every displayed v -dependent term, but leaves one integration constant, denoted temporarily by $B_2(\rho)$. The corner equation cannot determine this constant; it will be fixed by matching after the finite part of the outer second-order coefficient has been calculated. The L^2 term is the next-order continuation of the first-order logarithmic switchback.

We therefore return to the outer expansion and calculate that finite part.

Introduce the local distance from the final-size root,

$$s = x - x_f.$$

Define

$$f_1 := F'(x_f), \quad f_2 := F''(x_f).$$

Since

$$F'(x) = \frac{1 - \mathcal{R}_0 x}{\mathcal{R}_0 x}, \quad F''(x) = -\frac{1}{\mathcal{R}_0 x^2},$$

the definitions give

$$f_1 = F'(x_f) = \frac{\delta}{\mathcal{R}_0 x_f}, \quad f_2 = F''(x_f) = -\frac{1}{\mathcal{R}_0 x_f^2}.$$

From $G_\theta = \mathcal{R}_0 Y_1$, equations (49) and (55) identify the logarithmic coefficient and finite part as

$$\alpha = \frac{h_f}{\mathcal{R}_0 x_f}, \quad \beta = \alpha \log \frac{f_1}{C_\theta}.$$

To obtain the next local term, substitute $F(x) = f_1 s + \frac{1}{2} f_2 s^2 + O(s^3)$ and the Taylor expansions of $h_\theta(x)$ and x^{-2} into (40). This gives

$$Y'_1(x) = \frac{\alpha}{s} + p_0 + O(s), \quad p_0 = \frac{2\delta x_f h'_f - (1 + 2\delta)h_f}{2\mathcal{R}_0 \delta x_f^2}. \quad (66)$$

Integrating (66) and using the finite part β yields

$$Y_1(x) = \alpha \log s + \beta + p_0 s + O(s^2). \quad (67)$$

Define

$$A_{-1} = \frac{h_f/x_f - c}{\delta^2}, \quad B_0 = \beta - \alpha, \quad B_1 = -\frac{h_f}{2\mathcal{R}_0 \delta x_f^2}.$$

Denote the right-hand side of (41) by $q_2(x)$. Substituting (67) and $F(x) = f_1s + \frac{1}{2}f_2s^2 + O(s^3)$ into q_2 , then expanding its remaining factors, gives the part whose improper integral diverges at $s = 0$:

$$q_{2,\text{sing}}(s) = -\frac{a\alpha \log s + aB_0}{s^2} + \frac{A_{-1}\alpha \log s + A_{-1}B_0 - aB_1}{s}. \quad (68)$$

Define $q_{2,\text{reg}}(x) = q_2(x) - q_{2,\text{sing}}(x - x_f)$. A convenient explicit singular primitive is

$$P_2(s) = \frac{a\alpha \log s + a\beta}{s} + \frac{A_{-1}\alpha}{2} \log^2 s + (A_{-1}B_0 - aB_1) \log s, \quad (69)$$

for which $P_2' = q_{2,\text{reg}}$. Since $q_{2,\text{reg}}(x) = O((\log(x - x_f)))$ is integrable at x_f ,

$$K_2 = -P_2(1 - x_f) - \int_{x_f}^1 q_{2,\text{reg}}(u) du, \quad Y_2(x) = P_2(x - x_f) + K_2 + o(1). \quad (70)$$

The boundary condition $Y_2(1) = 0$, together with $q_2 = P_2' + q_{2,\text{reg}}$, gives (70) directly.

Substitute (67) and (70) into the inverse coefficient formula (57). Writing its numerator as

$$N_2(s) = \frac{1}{2}F''(x)X_1^2 + Y_1'(x)X_1 + Y_2(x), \quad (71)$$

and collecting the logarithmic and constant terms gives

$$N_2(s) = -f_1\kappa \left[(\log s)^2 + 2\frac{\beta}{\alpha} \log s \right] + N_{20} + o(1), \quad (72)$$

where the remaining constant term is

$$N_{20} = K_2 + \frac{\alpha\beta f_2}{f_1^2} - \frac{(\alpha + \beta)p_0}{f_1} + \frac{f_2\beta^2}{2f_1^2}. \quad (73)$$

Because $X_2 = -N_2/F'$ in (57), define

$$D_\theta = -\frac{N_{20}}{f_1} - \kappa \frac{\beta^2}{\alpha^2}. \quad (74)$$

Substitute (72) into (57). Then $y = f_1s + O(s^2)$ and $\beta/\alpha = \log(f_1/C_\theta)$ rewrite the result as

$$X_2(y) = \kappa \log^2 \left(\frac{C_\theta}{y} \right) + D_\theta + o(1) \quad (y \downarrow 0). \quad (75)$$

We next match this outer result to the corner solution and thereby determine the undetermined constant $B_2(\rho)$ in (65). In the overlap $\rho \ll y \ll 1$, set $y = \rho v$ and retain $L = \log(C_\theta/y) = \log(C_\theta/(\rho v))$. Taylor inversion of $y = F(x_f + s) = f_1s + \frac{1}{2}f_2s^2 + O(s^3)$ gives

$$X_0(y) - x_f = by + \frac{b\mathcal{R}_0}{2\delta^2}y^2 + O(y^3).$$

Moreover, substitution of $s = by + O(y^2)$ into the local formula for X_1 above, followed by collection of the constant and linear terms in y , gives

$$X_1(y) = aL + y \left[\frac{\mathcal{R}_0 a}{\delta^2}(L + 1) - \frac{bc}{\delta^2} \right] + O(y^2 \log y).$$

Using these two formulas together with (75) in $x = X_0 + \rho X_1 + \rho^2 X_2 + \dots$, and then substituting $y = \rho v$, yields

$$\begin{aligned} x - x_f &= \rho(aL + bv) \\ &+ \rho^2 \left[\kappa L^2 + D_\theta + \frac{\mathcal{R}_0 a}{\delta^2}(L + 1)v - \frac{bc}{\delta^2}v + \frac{b\mathcal{R}_0}{2\delta^2}v^2 \right] + o(\rho^2). \end{aligned}$$

The corner expansion is $x - x_f = \rho u_1(v) + \rho^2 u_2(v) + \dots$, with $u_1 = aL + bv$ and u_2 given by (65). All v -dependent terms already agree, and equality of the remaining constant terms gives $B_2(\rho) = D_\theta$. Therefore the matched second corner coefficient is

$$u_2(v) = \kappa L^2 + D_\theta + \frac{\mathcal{R}_0 a}{\delta^2}(L + 1)v - \frac{bc}{\delta^2}v + \frac{b\mathcal{R}_0}{2\delta^2}v^2. \quad (76)$$

9 Action formulation: from the corner expansion to a stable scalar equation

9.1 The action primitive

Define

$$\mathcal{A}'_{\theta}(x) = \frac{1 - \mathcal{R}_0 x}{h_{\theta}(x)} = \frac{1 - \mathcal{R}_0 x}{(1 - x)x^{\theta}}. \quad (77)$$

In the x -axis basin, where $y \ll \rho$, infection-induced depletion of susceptible individuals is of higher order, so

$$\frac{dx}{d\sigma} \simeq \rho h_{\theta}(x), \quad \frac{d \log y}{dx} = -\frac{1}{\rho} \mathcal{A}'_{\theta}(x).$$

Thus $\mathcal{A}_{\theta}$ is the natural variable linking deterministic epidemic matching to the later branching-process exponent.

For $\theta \neq 1$, one may take

$$\mathcal{A}_{\theta}(x) = \frac{x^{1-\theta}}{1-\theta} [\mathcal{R}_0 - (\mathcal{R}_0 - 1) {}_2F_1(1, 1 - \theta; 2 - \theta; x)]. \quad (78)$$

Important special cases are

$$\mathcal{A}_0(x) = \mathcal{R}_0 x + (\mathcal{R}_0 - 1) \log(1 - x), \quad (79)$$

$$\mathcal{A}_{1/2}(x) = 2\mathcal{R}_0 \sqrt{x} - 2(\mathcal{R}_0 - 1) \operatorname{artanh}(\sqrt{x}), \quad (80)$$

$$\mathcal{A}_1(x) = \log x + (\mathcal{R}_0 - 1) \log(1 - x). \quad (81)$$

9.2 Refined first-order action equation

Rather than truncating x directly, we reinsert (63) into the full action. This gives the practical refined first-order equation

$$\mathcal{A}_{\theta}(x_{\text{in}}^{(1)}) - \mathcal{A}_{\theta}(x_f) = \rho L_{\text{BL}} + \frac{b}{a} y_{\text{BL}}, \quad L_{\text{BL}} = \log \frac{C_{\theta}}{y_{\text{BL}}}. \quad (82)$$

Because

$$\frac{b}{a} = \frac{\mathcal{R}_0 x_f}{h_f},$$

this is exactly the finite- y_{BL} first-order formula.

9.3 Second-order action-resummed master equation

Define

$$Q = \frac{\mathcal{R}_0 a - bc}{a\delta^2}. \quad (83)$$

Substitute $x = x_f + \rho u_1 + \rho^2 u_2$ into $\mathcal{A}_{\theta}(x)$ and retain all terms required to second order under the corner scaling. A key simplification is that the L^2 term in u_2 cancels the quadratic contribution from the Taylor expansion of $\mathcal{A}_{\theta}$. The result is

$$\begin{aligned} \mathcal{A}_{\theta}(x_{\text{in}}^{(2)}) - \mathcal{A}_{\theta}(x_f) &= \rho L_{\text{BL}} + \frac{b}{a} y_{\text{BL}} + \rho^2 \frac{D_{\theta}}{a} \\ &\quad + \rho y_{\text{BL}} Q (L_{\text{BL}} + 1) + \frac{bQ}{2a} y_{\text{BL}}^2. \end{aligned} \quad (84)$$

This is the recommended second-order master equation.

Equation (84) has two important properties. First, it is a systematic higher-order extension of (82):

$$M_2 = M_1 + \text{second-order corrections.}$$

Second, it remains only a scalar root-finding problem for $x_{\text{in}}^{(2)}$. For fixed $(\mathcal{R}_0, \theta)$, the constants C_θ and D_θ need to be precomputed only once.

It is also important that the practical cutoff satisfies $y_{\text{BL}} \ll \rho$ but does not generally satisfy $y_{\text{BL}} \ll \rho^2$. Therefore, if one wants uniform second-order accuracy at finite K , it is not consistent to retain only the pure $\rho^2 D_\theta$ term while discarding the ρy_{BL} or y_{BL}^2 terms.

10 The Kendall layer: from x_{in} to burnout probability

Once the trajectory enters the low-prevalence layer, treat the susceptible path as a deterministic background and neglect the feedback of the rare infection on x :

$$\frac{dx}{d\sigma} \simeq \rho h_\theta(x).$$

Each infectious lineage then has birth rate $\mathcal{R}_0 x(\sigma)$ and removal rate 1. Kendall's result for a time-inhomogeneous linear birth–death process gives the extinction probability of one lineage as

$$q_1 = \frac{\mathcal{I}}{1 + \mathcal{I}}, \quad (85)$$

where

$$\mathcal{I}(x_{\text{in}}) = \int_{x_{\text{in}}}^1 \frac{1}{\rho h_\theta(X)} \exp \left[\frac{\mathcal{A}_\theta(X) - \mathcal{A}_\theta(x_{\text{in}})}{\rho} \right] dX. \quad (86)$$

If the matching line contains

$$m = K y_{\text{BL}}$$

infectious individuals, then the conditional probability of burnout is

$$P_{\text{burn}} \simeq \left(\frac{\mathcal{I}}{1 + \mathcal{I}} \right)^{K y_{\text{BL}}}, \quad (87)$$

and the probability of persistence through the first trough is

$$P_1 \simeq 1 - P_{\text{burn}}. \quad (88)$$

A numerically more stable quantity is the burnout exponent

$$B = -\log P_{\text{burn}} = K y_{\text{BL}} \log \left(1 + \frac{1}{\mathcal{I}} \right). \quad (89)$$

Hence the high-accuracy semi-analytical pipeline is

$$(\mathcal{R}_0, \theta) \rightarrow (x_f, C_\theta, D_\theta), \quad (\rho, K) \rightarrow y_{\text{BL}}, \quad (84) \rightarrow x_{\text{in}}^{(2)}, \quad (86) \rightarrow B, P_1.$$

The entire procedure requires only one-dimensional coefficient equations, a scalar root solve, and one one-dimensional Kendall integral; it does not require integrating the full two-dimensional epidemic ODE for each parameter set.

11 The Laplace layer: a fully asymptotic closed form

Define the action

$$\Lambda(x_{\text{in}}) = \frac{\mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_{\text{in}})}{\rho}. \quad (90)$$

The stationary point of the exponent in the Kendall integrand is determined by

$$1 - \mathcal{R}_0 x = 0,$$

so for every θ the unique internal saddle lies at

$$x_* = \frac{1}{\mathcal{R}_0}.$$

Applying Laplace's method around x_* gives

$$\mathcal{I} \sim \sqrt{\frac{2\pi}{\mathcal{R}_0 g(x_*)}} e^\Lambda, \quad g(x_*) = \rho h_\theta(x_*) = y^*. \quad (91)$$

The next coefficient follows by an independent local saddle expansion. Put $z = X - x_*$, $h_* = h_\theta(x_*)$, and $\lambda = \mathcal{R}_0/h_* > 0$. Since

$$\mathcal{A}'_\theta(x_* + z) = -\frac{\mathcal{R}_0 z}{h_\theta(x_* + z)},$$

the derivatives needed at the saddle are

$$\mathcal{A}'''_\theta(x_*) = \frac{2\mathcal{R}_0 h'_*}{h_*^2}, \quad \mathcal{A}''''_\theta(x_*) = -\frac{6\mathcal{R}_0 (h'_*)^2}{h_*^3} + \frac{3\mathcal{R}_0 h''_*}{h_*^2}.$$

For the amplitude $f = 1/h_\theta$, Gaussian averaging after $z = \sqrt{\rho} t$ gives the relative correction

$$\rho \left[\frac{f''_*}{2f_* \lambda} + \frac{f'_* \mathcal{A}''''_\theta(x_*)}{2f_* \lambda^2} + \frac{\mathcal{A}''''_\theta(x_*)}{8\lambda^2} + \frac{5\{\mathcal{A}'''_\theta(x_*)\}^2}{24\lambda^3} \right]. \quad (92)$$

Substitution and simplification yield

$$\mathcal{I} = e^\Lambda \sqrt{\frac{2\pi}{\mathcal{R}_0 \rho h_*}} \{1 + \rho c_L + O(\rho^2)\}, \quad c_L = \frac{h_*}{\mathcal{R}_0} \left[\frac{1}{12} \left(\frac{h'_*}{h_*} \right)^2 - \frac{1}{8} \frac{h''_*}{h_*} \right]. \quad (93)$$

This calculation verifies both the sign and normalization of the coefficient. For $h_\theta(x) = (1-x)x^\theta$,

$$h''_\theta(x) = \theta x^{\theta-2} \{(\theta-1) - (\theta+1)x\},$$

and in particular

$$c_L(\mathcal{R}_0, 0) = \frac{1}{12(\mathcal{R}_0 - 1)}. \quad (94)$$

Substituting this approximation into (89) gives the finite- m Laplace version; if in addition $\mathcal{I} \gg 1$, then

$$B \sim K y_{\text{BL}} \sqrt{\frac{\mathcal{R}_0 g(x_*)}{2\pi}} e^{-\Lambda}. \quad (95)$$

The generalized trough heuristic of Todd, David, and collaborators writes the same local saddle contribution in terms of the curvature at the trough. In the present model

$$\rho R'_{\text{eff}}(\tau_{\text{min}}) = \mathcal{R}_0 y^*,$$

so their expression reduces exactly to

$$B \sim K y_{\min} \sqrt{\frac{\mathcal{R}_0 y^*}{2\pi}}. \quad (96)$$

Strict-overlap matching makes this result independent of the matching line. At leading order,

$$\mathcal{A}_\theta(x_{\text{in}}) - \mathcal{A}_\theta(x_f) \sim \rho \log(C_\theta/y_{\text{BL}}), \quad y_{\min} = y_{\text{BL}} \exp\left[-\frac{\mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_{\text{in}})}{\rho}\right],$$

and hence

$$y_{\min}^{\text{BI}} \sim C_\theta e^{-\Delta A_f/\rho}, \quad B_{\text{BI}} \sim K C_\theta \sqrt{\frac{\mathcal{R}_0 y^*}{2\pi}} e^{-\Delta A_f/\rho}. \quad (97)$$

This BI formula is retained alongside, rather than substituted for, the finite-boundary exact-Kendall and Laplace approximations.

For the next-order BI approximation use the strict overlap $1/K \ll y_{\text{BL}} \ll \rho^2$. The second-order matching equation (84) then reduces to

$$\mathcal{A}_\theta(x_{\text{in}}) - \mathcal{A}_\theta(x_f) = \rho \log(C_\theta/y_{\text{BL}}) + \rho^2 D_\theta/a + o(\rho^2).$$

It follows directly that

$$\begin{aligned} y_{\min}^{\text{BI},2} &= y_{\text{BL}} \exp\left[-\frac{\Delta A_f}{\rho} + \log \frac{C_\theta}{y_{\text{BL}}} + \rho \frac{D_\theta}{a} + o(\rho)\right] \\ &= C_\theta \exp\left[-\frac{\Delta A_f}{\rho} + \rho \frac{D_\theta}{a} + o(\rho)\right]. \end{aligned} \quad (98)$$

Thus the artificial y_{BL} cancels before any probability is formed. Combining this matching correction with the reciprocal of the Kendall factor $1 + \rho c_L$ in (93) gives

$$B_{\text{BI}}^{\text{next}} = K C_\theta \sqrt{\frac{\mathcal{R}_0 y^*}{2\pi}} \exp\left[-\frac{\Delta A_f}{\rho} + \rho \left(\frac{D_\theta}{a} - c_L\right)\right], \quad P_{\text{persist}|\text{est}} = 1 - \exp(-B_{\text{BI}}^{\text{next}}). \quad (99)$$

Replacing $(1 + \rho c_L)^{-1}$ by $e^{-\rho c_L}$ changes only terms beyond the order retained. Equation (99) is therefore called a *next-order boundary-independent approximation*; it is not claimed to be a uniform second-order theory.

The action-resummed formulation has a further computational advantage. Let M_2 denote the right-hand side of (84), and define

$$\Delta A_f = \mathcal{A}_\theta(x_*) - \mathcal{A}_\theta(x_f).$$

Then one need not first solve for $x_{\text{in}}^{(2)}$ in order to obtain

$$\Lambda^{(2)} = \frac{\Delta A_f - M_2}{\rho}. \quad (100)$$

Thus the fully asymptotic prediction for P_1 can be computed directly from the matching constants and the action. The root solve for $x_{\text{in}}^{(2)}$ is required only when one wants the exact Kendall integral.

12 Why both exact Kendall and Laplace versions should be retained

Increasing the asymptotic order of x_{in} controls only the deterministic matching error; it cannot repair the saddle-point approximation itself. When

$$\Lambda \gg 1,$$

the Kendall integrand is sharply concentrated near x_* and the Laplace formula is accurate. When $\Lambda = O(1)$ or smaller, the Laplace approximation may have substantial error even if x_{in} is already extremely accurate.

The two versions should therefore be viewed as different levels of approximation:

1. **Primary numerical/semi-analytical prediction:** second-order x_{in} plus the exact one-dimensional Kendall integral;
2. **Fully asymptotic closed form:** second-order action plus Laplace’s method, valid when the action is sufficiently large.

Numerical validation should report “matching error” and “Laplace-only error” separately.

13 Domain of validity and identifiable failure modes

The present theory addresses deterministic-entry asymptotics for fixed $\mathcal{R}_0 > 1$, small ρ , and large K . Outside that regime, several distinct failure mechanisms can occur.

13.1 No boundary-layer entry

If the first deterministic trough satisfies

$$y_{\min} > y_{\text{BL}},$$

the descending branch of the first epidemic wave never reaches the matching line, so the present definition of x_{in} does not apply. Such parameter values should be labeled *no boundary-layer entry*, rather than assigned a large approximation error.

13.2 $\mathcal{R}_0 \rightarrow 1^+$

When $\mathcal{R}_0$ is very close to one, the first epidemic wave becomes weak and the scales associated with x_f , x_* , the peak, and the trough begin to merge. The fixed- $\mathcal{R}_0$ outer/corner hierarchy is then no longer uniform; a near-threshold distinguished limit is required.

13.3 Small-action Laplace breakdown

Even with an accurate x_{in} , Laplace’s approximation may fail if Λ is not large. In that regime the correct response is to retain the exact Kendall integral, rather than to increase the formal order of the x_{in} expansion.

13.4 Joint high- $\mathcal{R}_0$ limit

If $\mathcal{R}_0$ also becomes large so that x_f competes with ρ or smaller scales, a new boundary layer is required. Since

$$h_\theta(x) \sim x^\theta, \quad x \rightarrow 0,$$

the local balance between replenishment and infection-induced consumption,

$$\rho x^\theta \sim x,$$

gives the natural scale

$$x \sim \rho^{1/(1-\theta)}, \tag{101}$$

for $0 \leq \theta < 1$. The case $\theta = 1$ corresponds to a different distinguished limit. This is a future high- $\mathcal{R}_0$ uniformization problem, not simply an omitted ordinary $O(\rho^3)$ term in (84).

13.5 Finite- K stochastic entry

The present derivation treats x_{in} as deterministic. For finite K , the actual entry point is a random variable X_{in} , and its fluctuations may be exponentially amplified because

$$\Lambda'(x) = -\frac{1 - \mathcal{R}_0 x}{\rho h_\theta(x)}.$$

This effect is distinct from deterministic matching error and should be assessed separately through the distribution of entry points in stochastic simulations.

14 Theoretical chain and computational interface

At this stage the large-population deterministic-entry problem forms a closed chain:

$$(\mathcal{R}_0, \rho, \theta, K) \longrightarrow (x_f, C_\theta, D_\theta, y_{\text{BL}}) \longrightarrow x_{\text{in}}^{(1)}, x_{\text{in}}^{(2)} \longrightarrow \mathcal{I}, \Lambda \longrightarrow P_{\text{burn}}, P_1.$$

A practical implementation naturally separates into a precomputation layer and a parameter layer:

1. For each $(\mathcal{R}_0, \theta)$, compute x_f , C_θ , and D_θ , using the regularized quadrature (74) for the latter;
2. for each (ρ, K) , compute y^* and y_{BL} , and use (82) and (84) to obtain first- and second-order predictions;
3. for the high-accuracy version, use the exact Kendall integral (86);
4. when a closed form is needed, use the action (100) and the Laplace approximation (91);
5. record no-entry status, Λ , and the difference between first and second order as validity diagnostics.

Systematic numerical validation can compare the first- and second-order approximations with the deterministic ODE benchmark, exact Kendall integral, Laplace approximation, and full stochastic simulations, thereby mapping the domain of validity quantitatively.

15 Extension to population-size-dependent transmission

This section gives a separate derivation for Ben's count-level incidence. It does not alter any formula in the preceding $\psi = 0$ theory. The asymptotic limit throughout is

$$\rho \rightarrow 0, \quad \mathcal{R}_0 > 1, \quad \theta \in [0, 1], \quad \psi \in [0, 1) \quad \text{fixed.}$$

Write $\eta = 1 - \psi > 0$ and define every power by its positive real branch.

15.1 Dimensional model and normalization

Let S and I be counts, $N = S + I$, K the reference population size, γ the infected removal rate, and rK the dimensional scale of susceptible recruitment. The deterministic count equations corresponding to incidence $(\beta/K)SI(K/N)^\psi$ are

$$\begin{aligned} \frac{dS}{dt} &= rKh_\theta(S/K) - \frac{\beta}{K}SI \left(\frac{K}{S+I} \right)^\psi, \\ \frac{dI}{dt} &= \frac{\beta}{K}SI \left(\frac{K}{S+I} \right)^\psi - \gamma I. \end{aligned} \tag{102}$$

Set $x = S/K$, $y = I/K$, $\sigma = \gamma t$, $\rho = r/\gamma$, and $\mathcal{R}_0 = \beta/\gamma$. Since $S = Kx$, $I = Ky$, and $N/K = x + y$, division of the first equation by $K\gamma$ gives, line by line,

$$\frac{dx}{d\sigma} = \frac{r}{\gamma} h_\theta(x) - \frac{\beta}{\gamma} xy \left(\frac{1}{x+y} \right)^\psi = \rho h_\theta(x) - \frac{\mathcal{R}_0 xy}{(x+y)^\psi}.$$

The same calculation for I gives

$$\frac{dy}{d\sigma} = \frac{\beta}{\gamma} xy(x+y)^{-\psi} - y = \left[\frac{\mathcal{R}_0 x}{(x+y)^\psi} - 1 \right] y.$$

Thus K cancels from the normalized deterministic ODE because $N/K = x + y$; it remains the system-size parameter in stochastic matching. Here $\mathcal{R}_0$ is the reference reproduction parameter at $N = K$ near the naive state.

15.2 Zero-recruitment geometry and exact invariant

Put $n = x + y$. Adding the two equations at $\rho = 0$ cancels incidence and gives $dn/d\sigma = -y$. Moreover

$$\frac{dx}{dn} = \frac{-\mathcal{R}_0 xy n^{-\psi}}{-y} = \mathcal{R}_0 x n^{-\psi}.$$

Therefore $n^{-\psi} dn = dx/(\mathcal{R}_0 x)$. Integrating from the naive-wave normalization $(x, n) = (1, 1)$ to (x, n) yields

$$\frac{n^{1-\psi} - 1}{1 - \psi} = \frac{1}{\mathcal{R}_0} \log x, \quad n_0(x) = \left(1 + \frac{\eta}{\mathcal{R}_0} \log x \right)^{1/\eta}.$$

The descending epidemic trajectory is

$$F_\psi(x) = n_0(x) - x, \quad F_\psi(x_{f,\psi}) = 0, \quad (103)$$

where the second equation defines the nontrivial root. In general it is a scalar root problem, not an elementary Lambert– W expression. At $\psi = 0$, $n_0 = 1 + \mathcal{R}_0^{-1} \log x$ and hence $F_0 = 1 - x + \mathcal{R}_0^{-1} \log x$, exactly as above.

Differentiating $n_0^\eta = 1 + (\eta/\mathcal{R}_0) \log x$ gives

$$\begin{aligned} n'_0 &= \frac{n_0^\psi}{\mathcal{R}_0 x}, & F'_\psi &= \frac{n_0^\psi}{\mathcal{R}_0 x} - 1, \\ n''_0 &= \frac{\psi n_0^{2\psi-1}}{\mathcal{R}_0^2 x^2} - \frac{n_0^\psi}{\mathcal{R}_0 x^2}, & F''_\psi &= n''_0. \end{aligned} \quad (104)$$

Higher derivatives needed by the second-order subtraction are obtained by differentiating these displayed expressions; no numerical differentiation is required.

The same geometry is encoded by

$$H(x, y) = n^\eta - \frac{\eta}{\mathcal{R}_0} \log x.$$

Indeed, using $n' = \rho h_\theta(x) - y$ and $x' = \rho h_\theta(x) - \mathcal{R}_0 xy n^{-\psi}$,

$$\begin{aligned} \frac{dH}{d\sigma} &= \eta n^{-\psi} \{ \rho h_\theta(x) - y \} - \frac{\eta}{\mathcal{R}_0 x} \{ \rho h_\theta(x) - \mathcal{R}_0 xy n^{-\psi} \} \\ &= \eta \rho h_\theta(x) \left\{ n^{-\psi} - \frac{1}{\mathcal{R}_0 x} \right\}. \end{aligned} \quad (105)$$

The two terms proportional to $yn^{-\psi}$ cancel explicitly. Thus H is conserved at $\rho = 0$; for $\rho > 0$, equation (105) is an equivalent organization of the same perturbation theory and an independent check on the phase-plane coefficients below.

15.3 Threshold and endemic equilibrium

For $y/x \rightarrow 0$, $n = x\{1 + O(y/x)\}$ and the infected growth rate is $\mathcal{R}_0 x^{1-\psi} - 1 + O(y/x)$. Hence

$$x_{*,\psi} = \mathcal{R}_0^{-1/\eta}. \quad (106)$$

At a positive exact equilibrium, however,

$$y_e = \rho h_\theta(x_e), \quad \frac{\mathcal{R}_0 x_e}{(x_e + y_e)^\psi} = 1. \quad (107)$$

Put $x_e = x_{*,\psi} + \rho\xi + O(\rho^2)$ and $h_* = h_\theta(x_{*,\psi})$. Taking logarithms of the second condition and using $y_e = \rho h_* + O(\rho^2)$ gives

$$0 = \log \mathcal{R}_0 + \log x_e - \psi \log(x_e + y_e) = \rho \left\{ \frac{\xi}{x_{*,\psi}} - \psi \frac{\xi + h_*}{x_{*,\psi}} \right\} + O(\rho^2).$$

Consequently

$$x_e = x_{*,\psi} + \rho \frac{\psi h_*}{\eta} + O(\rho^2), \quad y_e = \rho h_* + \rho^2 h'_\theta(x_{*,\psi}) \frac{\psi h_*}{\eta} + O(\rho^3). \quad (108)$$

Thus $y_*^{(0)} = \rho h_*$ may consistently be used in $y_{BL} = \sqrt{y_*^{(0)}/K}$ at the retained leading stochastic-matching order. Replacing it by the exact y_e changes the height relatively by $O(\rho)$ and must be included if a claimed finite- K second-order result is sensitive to that convention.

15.4 Exact phase plane and the two outer coefficient equations

Division of the two normalized ODEs gives the exact equation

$$\frac{dy}{dx} = \frac{\{\mathcal{R}_0 x(x+y)^{-\psi} - 1\}y}{\rho h_\theta(x) - \mathcal{R}_0 xy(x+y)^{-\psi}}. \quad (109)$$

To expand every population-size factor, write

$$y = F_\psi + \rho Y_1 + \rho^2 Y_2 + O(\rho^3), \quad n = n_0 + \rho Y_1 + \rho^2 Y_2 + O(\rho^3).$$

The binomial expansion, through the required order, is

$$\begin{aligned} n^{-\psi} = n_0^{-\psi} & \left[1 - \rho \psi \frac{Y_1}{n_0} \right. \\ & \left. + \rho^2 \left\{ -\psi \frac{Y_2}{n_0} + \frac{\psi(\psi+1)}{2} \frac{Y_1^2}{n_0^2} \right\} \right] + O(\rho^3). \end{aligned}$$

Define, only to keep the collection readable,

$$\begin{aligned} p_0 &= \mathcal{R}_0 x n_0^{-\psi}, \\ p_1 &= -p_0 \psi Y_1 / n_0, \\ p_2 &= p_0 \left\{ -\psi Y_2 / n_0 + \frac{1}{2} \psi(\psi+1) Y_1^2 / n_0^2 \right\}. \end{aligned}$$

Cross-multiplying (109), the denominator and numerator have coefficients

$$\begin{aligned} d_0 &= -p_0 F_\psi, & a_0 &= (p_0 - 1) F_\psi, \\ d_1 &= h_\theta - p_0 Y_1 - p_1 F_\psi, & a_1 &= (p_0 - 1) Y_1 + p_1 F_\psi, \\ d_2 &= -p_0 Y_2 - p_1 Y_1 - p_2 F_\psi, & a_2 &= (p_0 - 1) Y_2 + p_1 Y_1 + p_2 F_\psi. \end{aligned}$$

Equating powers of ρ in $(d_0 + \rho d_1 + \rho^2 d_2)(F'_\psi + \rho Y'_1 + \rho^2 Y'_2) = a_0 + \rho a_1 + \rho^2 a_2$ gives, without omitted algebra,

$$d_0 F'_\psi = a_0,$$

$$d_0 Y'_1 + d_1 F'_\psi = a_1, \quad (110)$$

$$d_0 Y'_2 + d_1 Y'_1 + d_2 F'_\psi = a_2. \quad (111)$$

With $Y_1(1) = Y_2(1) = 0$, these are two sequential scalar coefficient equations. They are the production definition used below. When $\psi = 0$, $p_0 = \mathcal{R}_0 x$, $p_1 = p_2 = 0$; solving the last two displayed equations gives

$$Y'_1 = -\frac{h_\theta(x)(\mathcal{R}_0 x - 1)}{\mathcal{R}_0^2 x^2 F_0}, \quad Y'_2 = \frac{h_\theta(x)(\mathcal{R}_0 x - 1)}{\mathcal{R}_0^2 x^2 F_0^2} \left(Y_1 - \frac{h_\theta(x)}{\mathcal{R}_0 x} \right),$$

which are exactly the two equations in the validated theory. Expanding equation (105) in the same series reproduces equation (110); this direct-versus-invariant agreement checks the signs of all p_1 terms.

15.5 Finite-part matching constants

Let $x_f = x_{f,\psi}$ within this subsection, $s = x - x_f$, and $f_j = F_\psi^{(j)}(x_f)$. On the descending branch $f_1 > 0$. Expanding the right-hand side of (110) at fixed parameters determines coefficients $q_{1,-1}$ and $q_{1,0}$ by

$$Y'_1(x_f + s) = \frac{q_{1,-1}}{s} + q_{1,0} + O(s). \quad (112)$$

They are explicit rational combinations of $x_f, n_0(x_f) = x_f, f_1, f_2, h_\theta(x_f), h'_\theta(x_f), \mathcal{R}_0, \psi$ obtained by substitution in (110); this definition avoids unstable numerical differentiation. The finite regularized quadrature

$$K_1 = -q_{1,-1} \log(1 - x_f) - \int_{x_f}^1 \left\{ Y'_1(u) - \frac{q_{1,-1}}{u - x_f} \right\} du \quad (113)$$

gives $Y_1 = q_{1,-1} \log s + K_1 + q_{1,0} s + O(s^2)$. Implicit inversion of $y = F_\psi(x) + \rho Y_1(x) + \rho^2 Y_2(x) + \dots$ gives, term by term,

$$\begin{aligned} X_1 &= -Y_1/F'_\psi, \\ X_2 &= -\{\tfrac{1}{2} F''_\psi X_1^2 + Y'_1 X_1 + Y_2\}/F'_\psi. \end{aligned} \quad (114)$$

Since $s = y/f_1 + O(y^2)$, define

$$a_{\theta,\psi} = q_{1,-1}/f_1, \quad C_{\theta,\psi} = f_1 \exp(-K_1/q_{1,-1}). \quad (115)$$

Equations (112)–(115) then give $X_1(y) = a_{\theta,\psi} \log(C_{\theta,\psi}/y) + o(1)$. At $\psi = 0$, direct substitution gives $q_{1,-1} = h_\theta(x_f)/(\mathcal{R}_0 x_f)$ and equations (113)–(115) reduce algebraically to (47)–(55).

For second order, expand the fully explicit right-hand side of (111), after inserting (112), as

$$Y'_2(x_f + s) = \frac{r_{-2,1} \log s + r_{-2,0}}{s^2} + \frac{r_{-1,1} \log s + r_{-1,0}}{s} + O(\log s). \quad (116)$$

Every $r_{j,k}$ is fixed by coefficient extraction from displayed equations (110)–(111); hence it is explicitly calculable from the parameters, x_f , h_θ and their derivatives. An antiderivative of the four singular terms is

$$P_{2,\psi}(s) = -\frac{r_{-2,1}(\log s + 1) + r_{-2,0}}{s} + \frac{r_{-1,1}}{2} \log^2 s + r_{-1,0} \log s. \quad (117)$$

The finite, non-extrapolative production quadrature is therefore

$$K_{2,\psi} = -P_{2,\psi}(1 - x_f) - \int_{x_f}^1 \{Y_2'(u) - P_{2,\psi}'(u - x_f)\} du. \quad (118)$$

Substitution of $Y_2 = P_{2,\psi} + K_{2,\psi} + o(1)$ in (114), followed by $s = y/f_1 + O(y^2)$, uniquely produces

$$X_2(y) = \kappa_{\theta,\psi} \log^2 \frac{C_{\theta,\psi}}{y} + D_{\theta,\psi} + o(1). \quad (119)$$

Equation (119) defines $D_{\theta,\psi}$ as the constant selected by the local trajectory, while (118) computes it using only finite regularized one-dimensional quadratures. It is not a small- y extrapolation. For $\psi = 0$, the four coefficients in (116) reduce to the singular subtraction in (68), $P_{2,\psi}$ reduces to (69), and (114) reduces to (74). This is the exact symbolic regression check against the current theory.

15.6 Corner, direct entry, and action matching

Insert $y = \rho v$ and $x = x_f + \rho u_1(v) + \rho^2 u_2(v) + \dots$ directly in the inverse of (109). Taylor expansion of $\mathcal{R}_0 x(x + \rho v)^{-\psi}$ defines the positive local gap

$$\delta_\psi = 1 - \mathcal{R}_0 x_f^\eta, \quad h_f = h_\theta(x_f).$$

At leading order, division of numerator and denominator gives

$$u_1'(v) = -\frac{h_f}{\delta_\psi v} + \frac{\mathcal{R}_0 x_f^\eta}{\delta_\psi},$$

and matching to (115) fixes

$$u_1 = a_{\theta,\psi} L + b_\psi v, \quad L = \log \frac{C_{\theta,\psi}}{y}, \quad a_{\theta,\psi} = \frac{h_f}{\delta_\psi}, \quad b_\psi = \frac{\mathcal{R}_0 x_f^\eta}{\delta_\psi}. \quad (120)$$

This also checks $q_{1,-1}/f_1 = h_f/\delta_\psi$. At the next order the same quotient expansion yields a linear first-order equation $u_2' = G_{\theta,\psi}(v, u_1)$ whose coefficients are obtained by the displayed Taylor rule; its only integration constant is fixed by outer matching to be $D_{\theta,\psi}$. Retaining the complete finite-height solution of that equation, rather than only its L^2 term, is required before claiming a second-order direct entry formula. The verified first-order direct formula is

$$x_{\text{in}}^{(1),\text{dir}} = x_f + \rho a_{\theta,\psi} L_{\text{BL}} + b_\psi y_{\text{BL}}. \quad (121)$$

In the low-prevalence basin, $x' = \rho h_\theta(x) + o(\rho)$ and $(\log y)' = \mathcal{R}_0 x^\eta - 1 + o(1)$. Hence

$$\mathcal{A}'_{\theta,\psi}(x) = \frac{1 - \mathcal{R}_0 x^\eta}{h_\theta(x)}, \quad \frac{d \log y}{dx} = -\frac{\mathcal{A}'_{\theta,\psi}(x)}{\rho}. \quad (122)$$

Action differences are evaluated by the finite quadrature (1). Taylor expansion about x_f converts (121) to the resummed equation

$$\mathcal{A}_{\theta,\psi}(x_{\text{in}}^{(1)}) - \mathcal{A}_{\theta,\psi}(x_f) = \rho L_{\text{BL}} + \frac{b_\psi}{a_{\theta,\psi}} y_{\text{BL}} =: M_1. \quad (123)$$

At second order, the coefficient of L^2 in u_2 obtained from the corner ODE is exactly the negative of the L^2 coefficient generated by $\frac{1}{2} \mathcal{A}''(x_f)(\rho a L)^2$ after division by $\mathcal{A}'(x_f) = 1/a$. Thus the squared logarithm cancels in the action, as it does at $\psi = 0$. The remaining finite constant contributes $\rho^2 D_{\theta,\psi}/a_{\theta,\psi}$ to M_2 ; all $\rho y_{\text{BL}} L$, ρy_{BL} and y_{BL}^2 terms must be kept from the integrated u_2 equation. Admissibility is $0 < M_j < \Delta A_f$, where

$$\Delta A_f = \mathcal{A}_{\theta,\psi}(x_{*,\psi}) - \mathcal{A}_{\theta,\psi}(x_f).$$

15.7 Kendall, Laplace, and boundary-independent limit

For a rare infected individual the count incidence divided by I and by γ is $b(x) = \mathcal{R}_0 x(x+y)^{-\psi} = \mathcal{R}_0 x^\eta + O(y/x)$; removal gives $d(x) = 1$. At the naive state $x+y=1$, $x=1$, so $b = \mathcal{R}_0$ and the establishment probability remains $1 - 1/\mathcal{R}_0$. Kendall's formula and $d\sigma = dx/\{\rho h_\theta(x)\}$ give exactly (2). With $m = Ky_{\text{BL}}$,

$$P_{\text{burn|est}} = \left(\frac{\mathcal{I}}{1 + \mathcal{I}} \right)^m, \quad B = -\log P_{\text{burn|est}} = m \log(1 + 1/\mathcal{I}),$$

which is evaluated from $\log \mathcal{I}$ using 'log1p' arithmetic to avoid overflow.

The saddle satisfies $\mathcal{A}' = 0$, hence $x = x_{*,\psi}$. Differentiating (122) and using $\mathcal{R}_0 x_*^\eta = 1$ gives

$$\mathcal{A}_{\theta,\psi}''(x_*) = -\frac{\mathcal{R}_0 \eta x_*^{\eta-1}}{h_\theta(x_*)} = -\frac{\eta}{x_* h_\theta(x_*)}. \quad (124)$$

Laplace's method therefore yields

$$\mathcal{I} \sim \exp(\Lambda_j) \sqrt{\frac{2\pi x_*}{\eta \rho h_\theta(x_*)}}, \quad \Lambda_j = \frac{\Delta A_f - M_j}{\rho}. \quad (125)$$

This curvature calculation, rather than a formal replacement in the old formula, supplies the characteristic factor $\eta^{-1/2}$. Equations (2) and (125), each evaluated with first- or second-order entry, define four distinct approximations.

In the strict first-order overlap, finite-height terms vanish because $y_{\text{BL}}/\rho \rightarrow 0$, whereas $M_1/\rho = L_{\text{BL}} + o(1)$. Therefore

$$\begin{aligned} B_{\text{BI}}^{(1)} &\sim Ky_{\text{BL}} \sqrt{\frac{\eta \rho h_\theta(x_*)}{2\pi x_*}} \exp\{-\Delta A_f/\rho + L_{\text{BL}}\} \\ &= KC_{\theta,\psi} \sqrt{\frac{\eta \rho h_\theta(x_*)}{2\pi x_*}} e^{-\Delta A_f/\rho}. \end{aligned}$$

In the stricter second-order overlap $y_{\text{BL}} \ll \rho^2$, the $\rho y_{\text{BL}} L$, ρy_{BL} , and y_{BL}^2 terms vanish, while $M_2/\rho = L_{\text{BL}} + \rho D_{\theta,\psi}/a_{\theta,\psi} + o(\rho)$. Thus the matching contribution is

$$B_{\text{BI}}^{(2)} \sim B_{\text{BI}}^{(1)} \exp\{\rho D_{\theta,\psi}/a_{\theta,\psi}\}, \quad (126)$$

before the separately calculable next Laplace coefficient is included. The same cancellation gives

$$y_{\text{min}} \sim C_{\theta,\psi} e^{-\Delta A_f/\rho} \exp\{\rho D_{\theta,\psi}/a_{\theta,\psi}\}.$$

Since the local increase of $R_{\text{eff}} = \mathcal{R}_0 x^\eta$ at the trough is $\rho \mathcal{R}_0 \eta x_*^{\eta-1} h_\theta(x_*) = \rho \eta h_\theta(x_*)/x_*$, this is precisely the generalized Todd–Earn trough form. Their heuristic supplies the branching/Laplace interpretation; equations (110)–(119) are the new ψ -dependent phase-plane matching derivation.

15.8 Validity, singular endpoint, and verification status

The construction requires fixed $\psi < 1$, $\rho \rightarrow 0$, fixed $\mathcal{R}_0 > 1$ and θ , a deterministic-entry overlap $1/K \ll y_{\text{BL}} \ll \rho$, and a large action for Laplace's approximation. It is not uniform as $\psi \rightarrow 1^-$. In particular $x_* = \mathcal{R}_0^{-1/\eta}$ and the nontrivial x_f approach the boundary exponentially, so one additionally needs the corner displacement and matching height small relative to both x_f and the gap $x_* - x_f$. At $\psi = 1$, $\eta = 0$, the threshold and final-size geometry change and the fixed- $\psi < 1$

corner/action scaling fails. The case $\psi = 1$ requires a separate asymptotic analysis and is left for future work.

The algebraic $\psi = 0$ checks have been displayed at each stage. The existing validation code concerns $\psi = 0$ only. Numerical ODE/order and shrinking-cutoff tests for $\psi = 0.25, 0.5, 0.75$ require an implementation of equations (110)–(118); no unperformed numerical check is claimed here. Likewise, a complete finite-height printed formula for u_2 has deliberately not been inferred from the $\psi = 0$ expression: it must be generated from the generalized corner quotient before being used in a production second-order entry calculation.

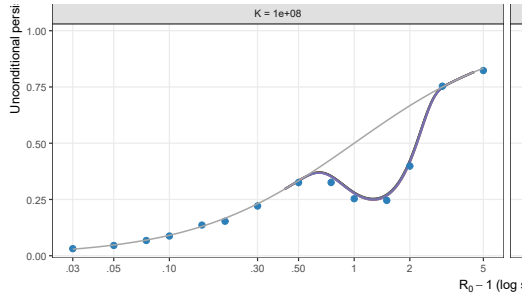
16 Numerical validation of the matching height

The matching height must be low enough for the local approximation to be accurate, but high enough to be reached by the deterministic first trough. If the line is too low, the present matching construction has no entry point. Raising it increases the parameter range with a usable crossing, but may reduce accuracy. The absence of a deterministic crossing does not imply that the probability of burnout during the first trough is negligible.

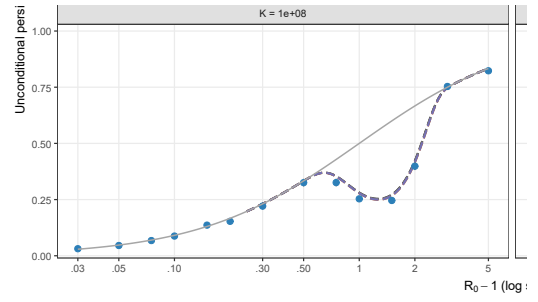
We compare four choices,

$$y_{\text{BL}} = \sqrt{\frac{y^*}{K}}, \quad y_{\text{BL}} = K^{-1/3}(y^*)^{2/3}, \quad y_{\text{BL}} = K^{-1/4}(y^*)^{3/4}, \quad y_{\text{BL}} = y^*.$$

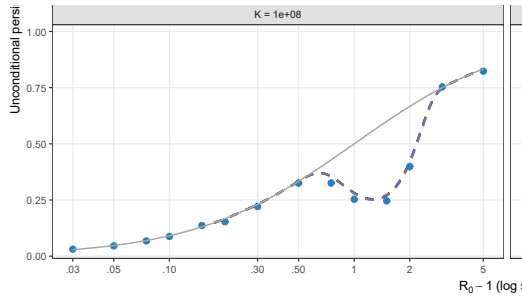
The middle two are the 2/3 and 3/4 compromises; $y_{\text{BL}} = y^*$ is the highest choice. Figures 1 and 2 show that higher boundary layers are less accurate than lower boundary layers. Figures 3 and 4 show that lower boundary layers have a narrower range of applicability than higher boundary layers.



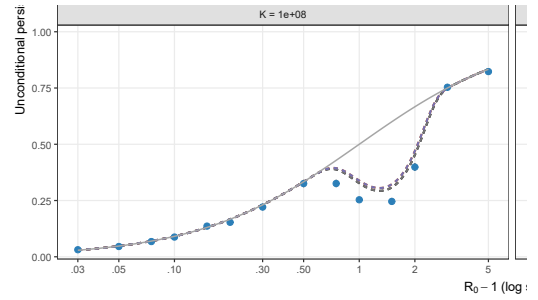
(a) $y_{BL} = \sqrt{y^*/K}$



(b) 2/3 compromise

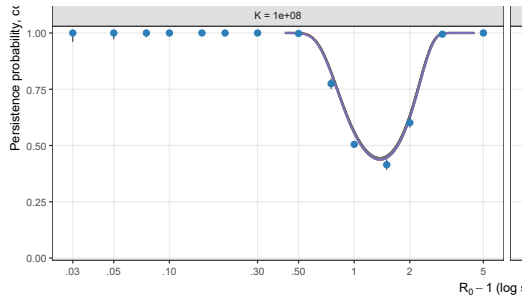


(c) 3/4 compromise

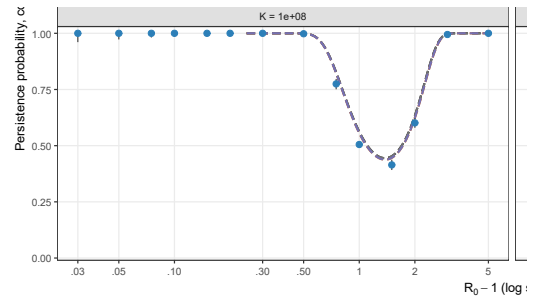


(d) $y_{BL} = y^*$

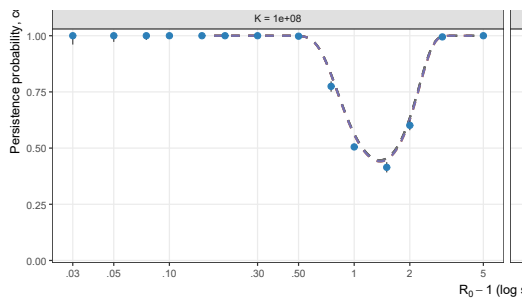
Figure 1: Unconditional persistence for $\rho = 0.01$, $\theta = 0$, and $K = 10^8$. Finite-boundary curves stop where no admissible analytical matching root exists; the BI curve remains defined there.



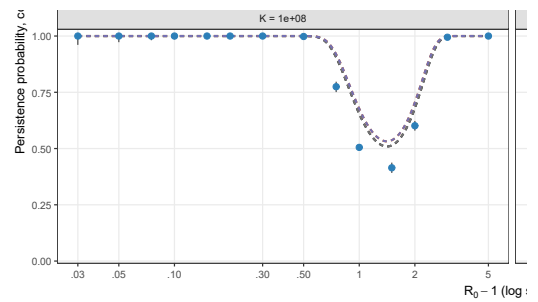
(a) $y_{BL} = \sqrt{y^*/K}$



(b) 2/3 compromise



(c) 3/4 compromise



(d) $y_{BL} = y^*$

Figure 2: Persistence conditional on escaping early fizzle for $\rho = 0.01$, $\theta = 0$, and $K = 10^8$. Finite-boundary curves stop where no admissible analytical matching root exists; the BI curve remains defined there.

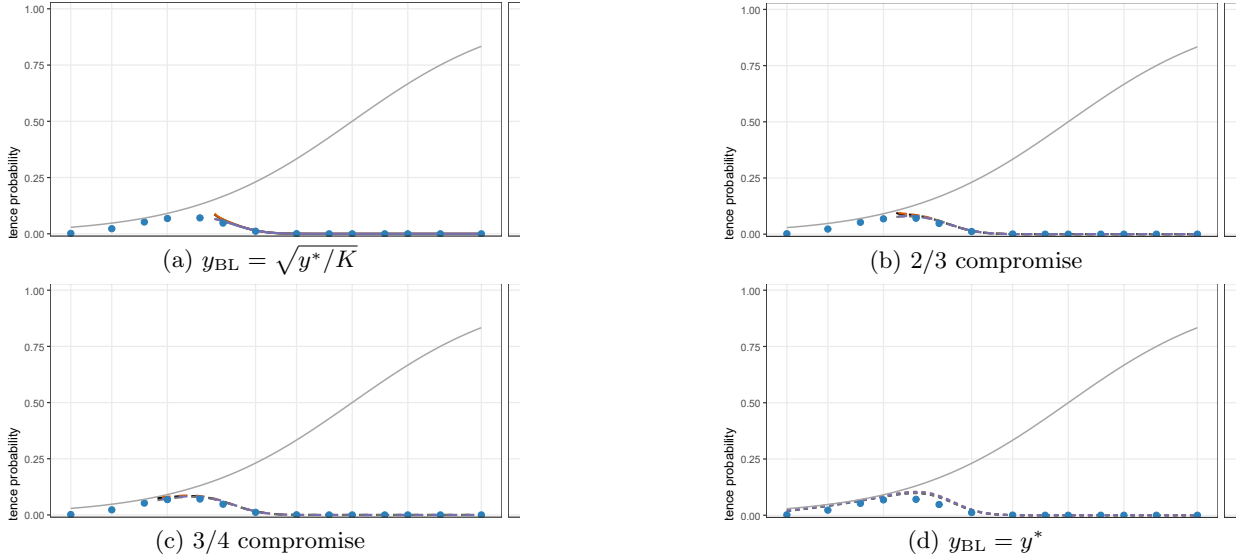


Figure 3: Unconditional persistence for $\rho = 0.01$, $\theta = 0.5$, and $K = 10^6$. Finite-boundary curves stop where no admissible analytical matching root exists; the BI curve remains defined there.

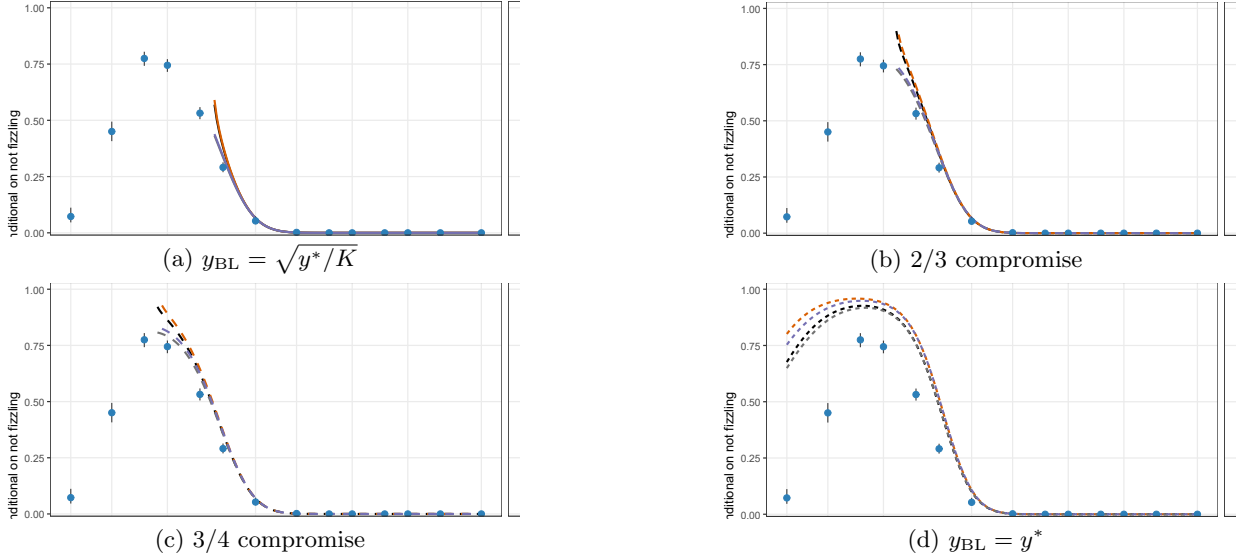


Figure 4: Persistence conditional on escaping early fizzle for $\rho = 0.01$, $\theta = 0.5$, and $K = 10^6$. Finite-boundary curves stop where no admissible analytical matching root exists; the BI curve remains defined there.

16.1 Boundary-layer-independent validation on the existing grid

The BI formula was evaluated on all 624 previously simulated parameter points; the stochastic counts, persistence definitions, and Wilson intervals were not recomputed. Its conditional mean absolute probability error is 0.0402 (median 0.0046, maximum 0.5615), while its unconditional mean absolute error is 0.0058 (median 0.0019, maximum 0.0523). On the 346 points where the finite-boundary construction has an entry, BI has conditional and unconditional mean absolute errors 0.0067 and 0.0026, respectively. On the 278 old NO_ENTRY points, which are all retained for BI, these errors rise to 0.0819 and 0.0097.

For comparison on their 345 finite predictions, second-order exact Kendall has conditional/unconditional mean absolute errors 0.0127/0.0052, and second-order finite-boundary Laplace has 0.0081/0.0030. Thus BI is slightly better on the common finite-entry subset, but

its all-grid conditional error is dominated by no-entry points and it should not be presented as uniformly more accurate. The clearest failures occur for slow replenishment no longer being strongly separated ($\rho = 0.10$), small $\mathcal{R}_0 - 1$, and moderate-to-large K ; there BI systematically underpredicts conditional persistence.

Because that conditional error is substantial, we also used the existing deterministic ODE trough cache as a secondary diagnostic. At $\rho = 0.02$ the median ratio $y_{\min}^{\text{BI}}/y_{\min}^{\text{ODE}}$ is close to one across θ . At $\rho = 0.10$ it falls from about 0.52 for $\theta = 0$ to 0.81 for $\theta = 1$, confirming that leading-order trough underprediction is an important cause of the persistence error in the weak-separation regime.

The next-order BI comparison uses exactly the same 624 stochastic points and Wilson intervals. Its all-grid conditional mean absolute error is 0.0346, compared with 0.0402 for leading BI. The combined correction improves 47.8% of individual points, so the lower mean error is not a uniform pointwise gain. For $\mathcal{R}_0 \leq 1.2$ the mean error falls from 0.0870 to 0.0730, driven mainly by the moderate- ρ points where the D_θ correction offsets leading underprediction; this empirical gain must not be mistaken for near-threshold asymptotic validity. For $\mathcal{R}_0 > 1.2$ the mean changes only from 0.0109 to 0.0107. At $\rho = 0.01$ the stochastic differences away from threshold are extremely small, while at $\rho = 0.10$ the correction can over-correct away from the leading regime.

The direct Kendall check separates the two ingredients. For $\mathcal{R}_0 > 1.2$ and $\rho = 0.01$, the mean absolute relative error of the leading saddle factor is 1.30×10^{-3} and that of the c_L -corrected factor is 6.81×10^{-6} ; the correction improves every such point. At $\mathcal{R}_0 \leq 1.2$, or at larger ρ , the next term is not uniformly smaller and can worsen the integral approximation, consistently with the known nonuniformity. The D_θ and c_L effects often have opposite signs; the matching correction is numerically larger in the weak-separation points, whereas the Laplace term provides the clean asymptotic improvement to the Kendall integral.

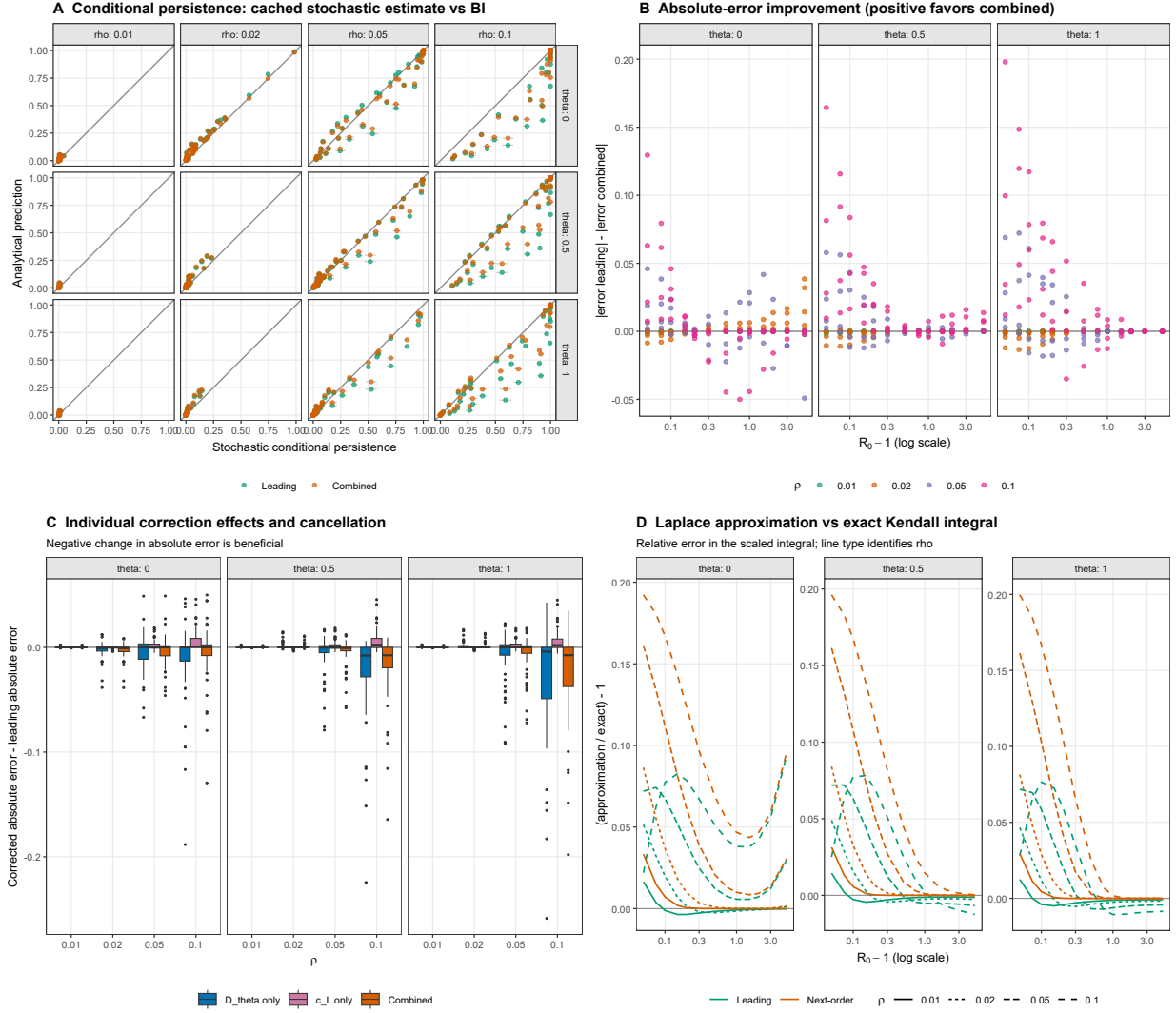


Figure 5: Cached-data validation of the next-order BI approximation. Panel A compares conditional stochastic persistence with leading and combined BI; panel B shows absolute-error improvement; panel C separates the D_θ and c_L effects; panel D compares leading and next-order Laplace factors directly with the exact Kendall integral. No stochastic simulations were rerun.

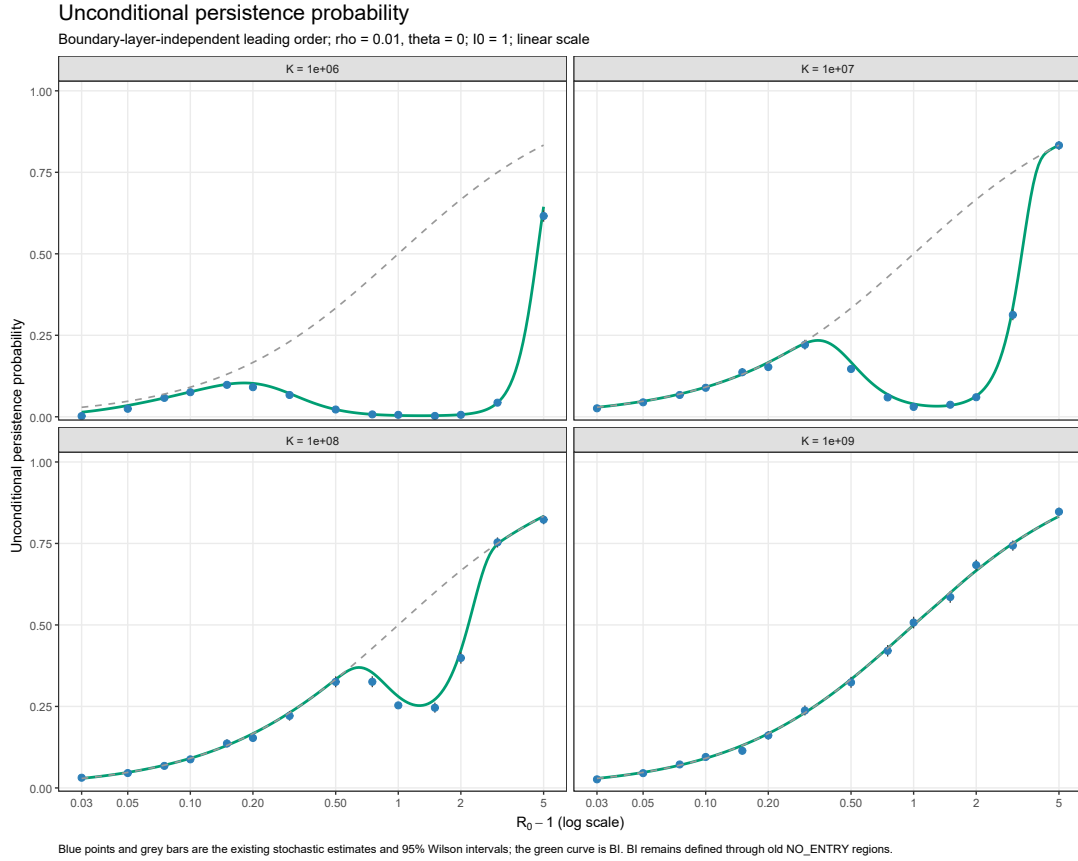


Figure 6: Unconditional BI prediction along $\mathcal{R}_0 - 1$ for the original Figure 11 large- K validation grid. Blue points and grey bars are the unchanged stochastic estimates and 95% Wilson intervals; the green BI curve remains defined through old NO_ENTRY regions. The accompanying multi-page PDF also contains the original Figure 9 grid.

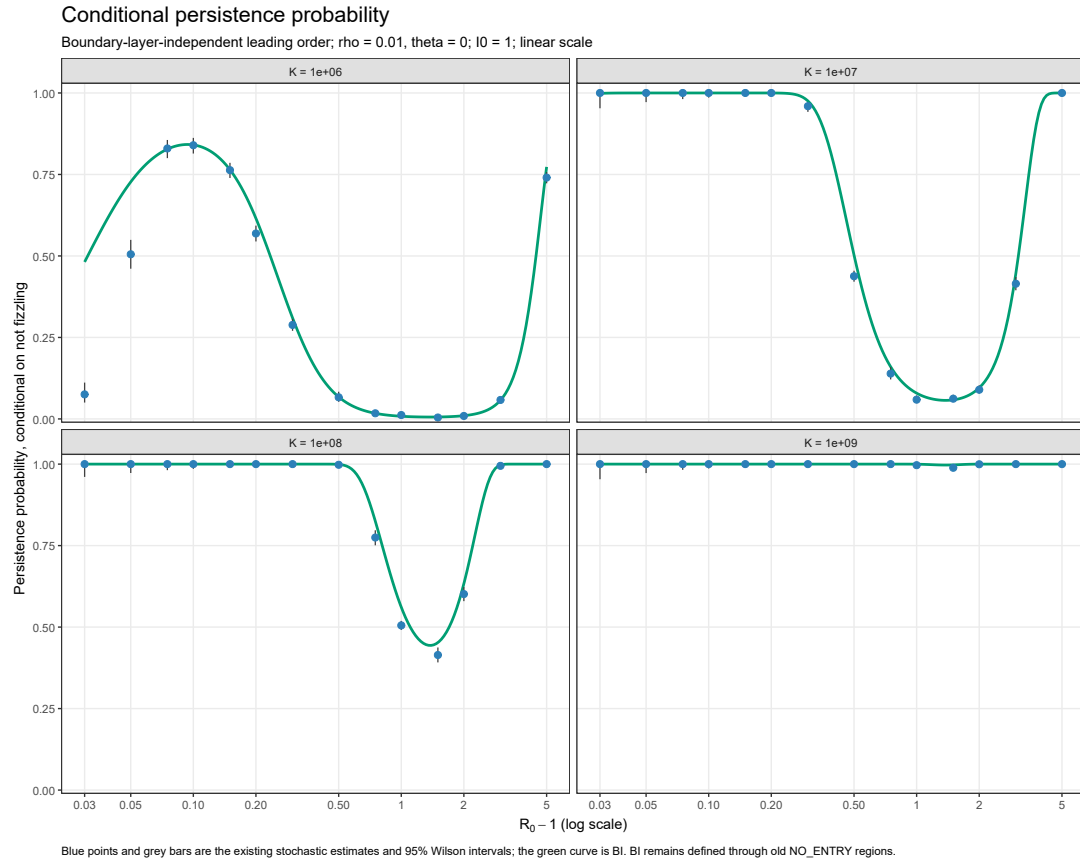


Figure 7: Conditional BI prediction on the same large- K grid. The accompanying multi-page PDF also contains the original Figure 10 grid.

A Core diagnostics to retain in numerical validation

To separate the main sources of approximation error, the numerical study should retain at least

$$\frac{y_{\min}}{y_{\text{BL}}}, \quad \frac{x_{\text{in}}^{(1)} - x_{\text{in}}^{\text{ODE}}}{x_{\text{in}}^{\text{ODE}}}, \quad \frac{x_{\text{in}}^{(2)} - x_{\text{in}}^{\text{ODE}}}{x_{\text{in}}^{\text{ODE}}},$$
$$\Delta\Lambda_1, \quad \Delta\Lambda_2, \quad \Lambda_{\text{ODE}}, \quad B_{\text{Kendall}}, \quad B_{\text{Laplace}}.$$

The regularized quadrature for D_θ should also be tested for endpoint-treatment and numerical-precision sensitivity.

References

- [1] T. L. Parsons and D. J. D. Earn, *Uniform Asymptotic Approximations for the Phase Plane Trajectories of the SIR Model with Vital Dynamics*, SIAM Journal on Applied Mathematics 84(4), 1580–1608 (2024).
- [2] T. L. Parsons, B. M. Bolker, J. Dushoff, and D. J. D. Earn, *The probability of epidemic burnout in the stochastic SIR model with vital dynamics*, Proceedings of the National Academy of Sciences 121(5), e2313708120 (2024).